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Macrocyclization of quinazoline-based EGFR inhibitors lead to exclusive mutant selectivity for EGFR L858R and Del19

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Author Contributions

J.A.A. and T.S.B. contributed equally to the work. J.A.A., S.K. and T.H. designed the project; J.A.A. and J.W. synthesized the compounds; T.S.B. performed the X-ray crystallography and docking experiments; T.S.B., I.K.S., J.K.R., and W.W.F. performed the biological assays; A.K. provided the proteins for the DSF assay; P.A.J., M.J.E. and S.K. supervised the research; the manuscript was written by J.A.A., S.K. and T.H. with contributions from all coauthors.

The authors declare no competing financial interest.

Associated content

Supporting Information Available: The supporting Information contains Figure S1-S6, Table S1-S8, and analytical data of compounds 2–18.

Molecular formula strings

Accession Codes

PDB ID codes: coordinates and structure factors of the EGFR-inhibitor complexes are available in the Protein Data Bank (PDB) under accession codes 7U99 (EGFR wt – **9d** complex), 7U9A (EGFR wt – **2f** complex), and 7U98 (EGFR T790M/V948R – **9f** complex).

Authors will release the atomic coordinates upon article publication.

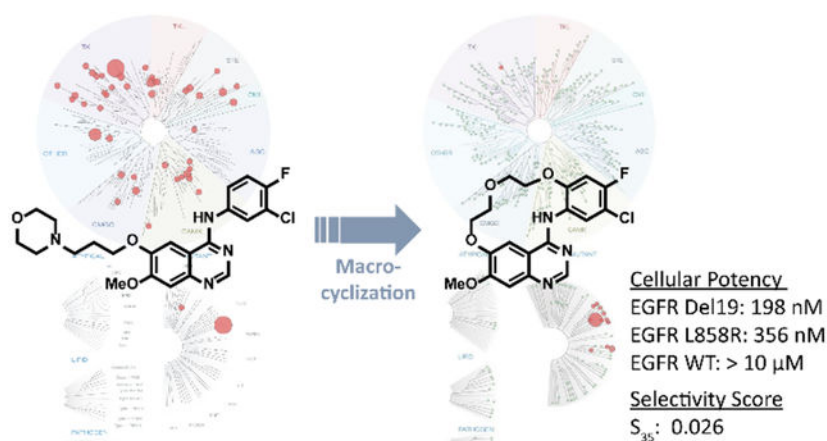
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Abstract

Activating mutations in epidermal growth factor receptor (EGFR) are frequent oncogenic drivers of non-small-cell lung cancer (NSCLC). The most frequent alterations in EGFR are short in-frame deletions in exon 19 (Del19) and the missense mutation L858R, which both lead to increased activity and sensitization of NSCLC to EGFR inhibition. The first approved EGFR inhibitors used for first-line treatment of NSCLC, gefitinib and erlotinib, are quinazoline-based. However, both inhibitors have several known off-targets and they also potently inhibit wild type (WT) EGFR, resulting in side effects. Here, we applied a macrocyclic strategy on a quinazoline-based scaffold as a proof of concept study with the goal increasing kinome-wide selectivity of this privileged inhibitor scaffold. Kinome-wide screens and SAR studies yielded in **3f**, a potent inhibitor for the most common EGFR mutation (EGFR Del19: 119 nM) with selectivity against the wild type receptor (EGFR: > 10 μ M) and the kinome.

Graphical Abstract



Introduction

Lung cancer is the second most common cancer in the western world with the highest number of cancer associated deaths.¹ The majority of lung cancers (85%), are non-small-cell lung cancers (NSCLC), a class of epithelial cancers that are relatively insensitive to chemotherapy.² A frequent driver in NSCLC are members of the erbB family of receptor tyrosine kinases, such as the epidermal growth factor receptor (EGFR), encoded by erbB-1 (HER1) which is frequently overexpressed or mutated in NSCLC.³ The EGFR protein harbors an N-terminal extracellular domain, a helical transmembrane domain, a juxtamembrane sequence, an intracellular kinase domain, and an α C-terminal domain.⁴

Monomeric EGFR is catalytically inactive, but the binding of diverse growth factors such as EGF (epidermal growth factor) or TGF α (transforming growth factor) stimulate EGFR dimerization resulting in receptor activation. Association in EGFR homodimers or heterodimers with other HER family members results in autophosphorylation at the C-terminal domain and initiation of the EGFR signaling pathways.^{5,6,7} A broad range of pathways are stimulated, including the Ras-Raf-MEK-MAPK pathway⁸, PI3K-Akt-mTOR pathway,⁹ and the JAK-STAT pathway resulting in growth-promoting and anti-apoptotic phenotypes¹⁰ (Figure S1A).¹¹ Approximately 10-20% of the caucasian and 40-60% of asian NSCLC patients exhibit somatic mutations of EGFR.¹² These mutations can be divided into different classes, such as single-base point mutations, in-frame deletion- and insertion mutations. The most common activation mutations are the EGFR L858R point mutation, which is detected in approximately 41% of all cases, and the EGFR Del19, which lacks the EGFR residues 746-750 encoded by exon 19 which is present in approximately 47% of all NSCLC cases. Both mutants lead to high sensitivity to EGFR TKIs (tyrosine kinase inhibitors) such as gefitinib or erlotinib.^{13,14} The remaining approximately 10% of NSCLC cancers that harbor EGFR mutations, contain a diversity of less common mutations including exon 20 (Ex20) and Ex19 insertions (Figure S1B).^{15,16}

Treatment of EGFR-mutated NSCLC comprises first-generation quinazoline-based EGFR inhibitors, such as gefitinib or erlotinib. Both inhibitors have demonstrated survival benefits compared to monotherapies with platinum-based or docetaxel chemotherapy.^{17,18} Second-generation inhibitors comprise a number of covalent quinazoline-based inhibitors, such as afatinib and dacomitinib, which are characterized by significantly improved kinome-wide selectivity profiles compared to first-generation inhibitors but they equipotently inhibit wild type EGFR (Figure S2).^{19,20} However, in 50-70% of the successfully treated patients, a second T790M point mutation in exon 20 occurs.^{21,22} This so-called gatekeeper mutation substitutes the threonine for a methionine, weakening potency of the first and second-generation TKIs by a resistance mechanism that is at least in part due to an increased affinity of T790M for ATP.²³ Osimertinib is one of the first FDA and EMA approved third-generation TKIs designed to target T790M mutated NSCLC, which similar to afatinib covalently targets C797.^{24,25} However, approximately 20–40% of NSCLC patients suffer relapse due to a C797S mutation, to which osimertinib can no longer covalently bind.^{26,27}

In 2019 Engelhardt *et al.* published a macrocyclic compound BI-4020, which has been developed for the treatment of this third EGFR mutation C797S and which preferentially binds to mutant EGFR.²⁸ Interestingly, a macrocyclization strategy was used to generate a highly potent and selective EGFR inhibitor. In this article, we report the development of new macrocyclic TKIs based on a quinazoline scaffold. Rigidification of gefitinib through macrocyclization led to a highly potent mutant selective EGFR inhibitor, with excellent kinome wide selectivity (Figure 1).

Results and Discussion

For the synthesis of the macrocycles **3a–f** the commercially available 4-chloro-7-methoxyquinazolin-6-ol (**1**) was used. Diverse 2-aminophenol derivatives were coupled using a nucleophilic substitution reaction. For macrocyclization a double Mitsunobu reaction

was carried out employing a diethylene glycol linker to obtain the final compounds **3a–f** in yields between 4 and 18% and E_{mac} values^{29,30} between 2.8 and 4.8 (Scheme 1). The poor yields of the macrocyclization can be attributed, among other things, to an incomplete reaction. We observed that the linker was attached through a Mitsunobu reaction once, however the second *in situ* Mitsunobu reaction failed. Another side product, which was detected was the precursor **2a–f** with 2 attached linker moieties. However, dimerization could be avoided through a high dilution of the reaction mixture.

In order to improve the yields of the macrocyclization step and the purification of the final compounds, an alternative synthetic route was established for subsequent compounds. Here the linker moieties 6-chlorohexan-1-ol (**L₁**) and 2-(2-(2-chloroethoxy)ethoxy)ethanol (**L₂**) were used to react with the hinge binding quinazoline (**1**) via a Mitsunobu reaction, using triphenylphosphine (TPP) and diisopropyl azodicarboxylate (DIAD). This step was followed by a nucleophilic substitution, using diverse 2-aminophenol derivatives to obtain compounds **6a–f** and **7a–f**. The macrocyclization was carried out by a nucleophilic substitution, using sodium hydride to obtain the final compounds **8a–f** and **9a–f** with yields between 6 and 100% and E_{mac} values between 2.6 and 6.3 (Scheme 2).

The macrocycles **12a–d** were also synthesized via this alternative three-step synthesis route. For the synthesis of these compounds a 2-(2-chloroethoxy)ethanol linker was reacted with **1**. In the following step different 3-aminophenol derivatives were used to introduce an alternative attachment point for the linker. The ring-closing reaction was again carried out via a nucleophilic substitution with yields between 8 and 30% and E_{mac} values between 3.0 and 4.7 (Scheme 3).

To evaluate the effect of the solvent-exposed residue at the C7 position of the quinazoline, macrocycle **18** was synthesized in a five-step synthetic route. The commercially available 6-hydroxyquinazolin-4(3*H*)-one (**13**) was acetylated, then chlorinated with thionyl chloride and deprotected with ammonia in methanol. The synthetic route published by Lyssikatos et. al. was modified to obtain **16** in an overall yield of 16%.³¹ The next steps were nucleophilic substitution using 2-amino-4-chloro-5-fluorophenol and macrocyclization by a double Mitsunobu reaction with diethylene glycol, TPP, and DIAD to obtain the final compound **18** with a yield of 5% and an E_{mac} value of 3.1 (Scheme 4).

The activity of all synthesized macrocycles **3a–f**, **8a–f**, **9a–f**, **12a–d**, **18**, as well as the acyclic control compound **2f**, toward EGFR WT and a number of selected mutants (EGFR Del19 and EGFR d747-752/P753S) were evaluated in a radiometric protein kinase assay (Reaction Biology). An inhibitor concentration of 1 μM was used in this initial assessment (Table 1). The macrocycles **3b** and **3f** showed an interesting profile as they potently inhibited EGFR Del19 (15% and 12% remaining activity) but had less activity on EGFR WT. Most other macrocycles from this series were either inactive or showed comparable activity on both wild type as well as the mutant EGFR. Macrocylic derivatives **8b–f** and **9b–f**, which harbored a longer linker showed no preferred inhibition of mutant EGFR or were inactive. The *para*-chloro derivatives **8d**, **9d**, and the chloro-fluoro derivatives **8f**, **9f** were the most potent inhibitors of this series. The macrocycles **12b–d**, which contained the 3-aminophenol moiety, showed no activity for EGFR WT or the tested mutants. Also **18**, which lacked

the methoxy group at the C7 position of the quinazoline moiety, was surprisingly inactive compared to **3f**, despite this relatively rather small change in the solvent region. The acyclic counterpart **2f** was highly potent against all tested EGFR variants.

To further determine the inhibitory potential of all compounds against the most common EGFR mutations, the IC₅₀ values were determined for EGFR WT, EGFR Del19, EGFR L858R, and EGFR L858R/T790M using a homogeneous time-resolved fluorescence (HTRF) assay. Gefitinib, BI-4020, and osimertinib were also assessed for comparison and as positive controls (Table 1). In general, the macrocycles with the shorter linker (**3a-f**, **8a-f**) were mainly active on the EGFR Del19 mutant. Especially **3a**, **3d**, **3f**, **8d**, **8e** and **8f** potently inhibited EGFR Del19 with IC₅₀ values between 20 nM and 236 nM and exhibited a weaker activity on L858R (IC₅₀ = 128 nM – 821 nM) and an inactivity against EGFR WT (IC₅₀ > 10 μM). Macrocycles containing the longer linker **9b-f** were potent on both kinases, the EGFR Del19 and L858R mutant with IC₅₀ values in a low nanomolar range. **9a** seemed to be mutant selective for L858R (IC₅₀ = 88 nM), whereas it was not inhibiting Del19 (IC₅₀ = 1874 nM). The exchanged attachment point for the macrocycles **12a-d** seemed to be not tolerated and led to the inactivity of the compounds. As expected for this scaffold, the additional gatekeeper mutation L858R/T790M led to a weaker potency against the quinazoline-based macrocycles.

Macrocyclization yielded in EGFR inhibitors with exclusive mutant selectivity

To obtain insights into the selectivity profile of the synthesized series of quinazoline-based macrocycles, we used a differential scanning fluorimetry (DSF) assay comprising more than a hundred kinase catalytic domains. This rapid and sensitive assay format measured the melting temperature (ΔT_m) in absence and presence of a compound by monitoring changes in fluorescence of a dye molecule caused by interaction with hydrophobic surfaces exposed in the denatured state.³² Compound binding stabilizes the protein as indicated by a positive ΔT_m relative to ligand-free protein. The assay panel contained known off-targets of the non-cyclic lead structures such as the kinases ABL1 and GAK. Staurosporine (**19**) was used as a positive control (Table S1-S4). In this assay, **2f** revealed several hits with a ΔT_m > 3°C including DYRK1, DYRK2, GAK, BMP2K, STK17A, and STK17B. This implied a less favorable kinome-wide selectivity profile as opposed to the macrocyclic counterpart **3f**. The macrocycle gave rise to a clean ΔT_m profile with only 3 kinases with modest ΔT_m > 3°C.

Based on the activity of **3f** for mutant EGFR, its considerably weaker activity for wild type EGFR, and its clean selectivity profile in our DSF panel (Figure 2A), we selected macrocycle **3f** for further profile using a KINOMEScan selectivity screening (Eurofins Scientific). In this platform, **3f** was screened against 468 human protein kinases and their mutants at a screening concentration of 1 μM (Figure 2B, Table S6). Gratifyingly, an excellent selectivity profile was observed in this comprehensive selectivity panel with a selectivity score (S₃₅) as low as 0.026 (Figure 2C). The profiling revealed that **3f** was weakly active on the EGFR WT and a potent binder of the EGFR mutants EGFR d746-750 (Del19), EGFR d747-752/ P753S, EGFR L858R, or EGFR d752-759 in addition to weaker interaction detected for other oncogenic mutants (FLT3 D835Y, FLT3 ITD D835V) (Figure 2D).

To gain insight into the binding mode, we solved the crystal structures of a representative macrocycle **9d** and a corresponding non-cyclic inhibitor **2f**, bound to EGFR WT (PDB: 7U99, 7U9A) as well as **9f** bound to T790M/V948R (PDB: 7U98) (Figure 3A, Figure S3). The Del19 mutation, unfortunately, introduces flexibility into the EGFR catalytic domains which precluded structure determination of this mutant so far. A superimposition of **2f** with gefitinib (Figure 3B) revealed the expected canonical binding mode of the quinazoline core structure. However, the aromatic ring was slightly rotated probably due to the introduction of the additional alcohol group used for cyclization which interacted with the D855 of the DFG motif. The similar binding mode of **2f** compared to gefitinib is also reflected in the assay data, where **2f** has a related activity profile for EGFR WT and the mutants. An overlay of the macrocycle **9d** with gefitinib (Figure 3C) also revealed a similar binding mode as observed for gefitinib. However, the pendant aromatic ring was slightly rotated in comparison to gefitinib and the linker of the macrocycle **9d** was facing towards the C-terminal lobe. Figure 3A demonstrated an excellent shape complementarity of **9d** with the EGFR active site. To gain insight into the binding mode of the highly related macrocycle **3f** harboring the shorter linker, we docked this inhibitor into the active conformation of EGFR WT. Figure 3D and Figure 3E show a superimposition and a comparison of the observed binding modes of **9d** and **3f**. Due to the shortening of the linker, molecule **3f** appears less flexible, resulting in a different predicted ring orientation. Whereas **2f** and **9d** can adopt the bioactive conformation of gefitinib relatively well, it can be seen in the superimposition of **3f** with **9d** that the pendant aromatic ring of **3f** can no longer assume the bioactive conformation. This suggests that truncation of the linker moiety was sufficient to weaken the binding affinity of **3f** to EGFR WT and optimally position this inhibitor in the binding pocket of the Del19 mutant.

Subsequently, the antiproliferative activity of the selected compounds **2f**, **3f**, **9d**, **9f**, **18** was determined together with the bench marking compounds gefitinib, BI-4020, mobocertinib, and osimertinib in cell lines that depended on the diverse EGFR mutants for survival. The Ba/F3 cell model was used by transducing these cells with EGFR WT, L858R, L858R/C797S, L858R/T790M, L858R/T790M/C797S, Del19, Del19/C797S, Del19/T790M, Del19/T790M/C797S, and Ex20 insertion mutants (Table 2, Figure 4, Figure S4). Overall, the acyclic compound **2f** showed a similar activity profile to gefitinib for the mutants studied, while the inactive compound **18** had no impact on the cell proliferation below 10 μ M, except for a minor effect at the L858R, Del19, and Del19/C797S variants. These observations correlated very well with the binding and enzyme inhibition data supporting the hypothesis that the methoxy group in position C7 is mandatory for activity on EGFR WT and for most EGFR mutants. **3f** did not inhibit EGFR WT growth, whereas it inhibited both L858R and L858R/C797S mutants in the submicromolar range, with IC₅₀ values of 385.6 and 749.6 nM, respectively. Intriguingly, **3f** was even more potent in cells transduced with the Del19 and Del19/C797S mutations than on the other mutants in this series, with IC₅₀ values of 197.5 and 147.9 nM, respectively, which is in good agreement with enzyme inhibition data. The other tested mutants were not significantly inhibited. **9d** and **9f** showed good efficacy with IC₅₀ values in the low one-to two-digit nanomolar range for EGFR WT, L858R, Del19, and the respective double mutation including C797S. Overall, compound **9d** represents the most potent compound within this series, even with a superior efficacy profile compared to gefitinib. This is again evidence that macrocyclization can be used to increase the potency

of the acyclic compound, provided that the bioactive conformation can be adopted. None of the quinazoline-based macrocycles significantly inhibited cell proliferation in the Ex20 insertion mutations and the gatekeeper mutation T790M, which is well in line with the other quinazoline-based FDA-approved EGFR inhibitors.

Analysis of the selected compounds by Western blotting confirmed the activity profiles of these compounds, using phosphorylation/activation of EGFR as markers (Figure 5 and Figure S5). While the gatekeeper mutant T790M and wild type EGFR were unaffected by **3f**, this macrocyclic inhibitor showed significant activity at L858R and excellent activity at the Del19 and Del19/C797S mutant EGFR. The reference inhibitors included in this analysis reproduced published results.³³ **2f**, **9d**, and **9f** showed inhibitory activity on phosphorylation of EGFR L858R, Del19, and the respective double mutation with C979S and a weak activity on EGFR WT. Once more, **18** showed no activity on all mutants tested as well as the wild type receptor.

Additionally, the efficacies of selected inhibitors were evaluated on perturbing downstream signalling pathways including the PI3K/AKT/mTOR and RAS/MAPK signalling. Inhibition of AKT and ERK1/2 following treatment with selected inhibitors was determined for EGFR WT and the most interesting mutants (Figure 6). The results showed a good concordance between inactivation of EGFR and its downstream targets. **2f**, **9d**, and **9f** showed significant attenuation of the phosphorylation of ERK and AKT in the L858R, L858R/C797S, Del19, and Del19/C797S mutants. As expected, only weak inhibition on EGFR WT was observed. **3f** exhibited again mutant selectivity as demonstrated in inhibition of the downstream targets on the tested EGFR mutants, whereby EGFR WT was not effected. **18** showed a negligible effect on all tested variants.

In addition, the chemical stability of **3f** was determined using a PBS buffer (pH 7.4) and 0.1 M HCl (pH 1.2) to make a preliminary assessment of its pharmacokinetic properties. **3f** was incubated at rt for a total of 184 minutes, and the amount of the residual compound was determined by high-performance liquid chromatography (HPLC) every 23 minutes. **3f** showed excellent chemical stability under acid conditions with more than 95% after 3 hours and also good stability at pH 7.4 above 80% (Figure S6, Table S7). Additionally, the solubility of **3f** was tested in different solvents (Table S8) by incubating **3f** at 37°C for 24 hours. The amount of solubilized compound was determined by HPLC. The solubility in neutral aqueous solutions is rather poor with 0.131 – 0.174 µg/mL, however the solubility was improved by adding 5% or 10% DMSO (0.283 – 0.294 µg/mL). The solubility was further increased by using 100% DMSO or a simulated gastric fluid (SGF) buffer³⁴ with 2.592 mg/mL and 42.802 µg/mL, respectively.

Discussion and Conclusion

Macrocyclization has been used as a strategy to increase on-target potency while diminishing the off-targets of acyclic analogues due to conformational restriction. Here, we initiated a proof of concept study on a quinazoline scaffold, which is widely used in kinase drug discovery. In this study, we chose gefitinib, an approved inhibitor with multiple off-targets, as a starting point to demonstrate that the macrocyclization strategy can lead to

the development of highly selective compounds. The quinazoline-based inhibitors of first- and second-generation TKIs generally show decreased activity at the gatekeeper mutant T790M, which sterically prevents them to bind to the ATP binding pocket, and furthermore, this mutation resulted in an increased affinity of this mutant for ATP. To address this issue, third generation of TKIs was based on a different hinge-binding scaffold have been developed. Osimertinib, a third-generation TKI with a 2-aminopyrimidine hinge binding moiety, binds covalently to C797 and is used as a first-line therapy in NSCLC to target the T790M mutant. However, these covalent inhibitors fail in the C797S mutant, which is present in 20-40% of NSCLC patients.

We synthesized a series of diverse quinazoline-based macrocycles with different linker length and established a small SAR on modifications in the hydrophobic back-pocket and the solvent exposed front-pocket. One challenge during the macrocyclization step was the implementation of the optimal linker length that allows the macrocycle sufficient flexibility in order to adopt a bioactive conformation for potent target interaction without adopting conformation that leads to off-target activity. To efficiently introduce linker variations, we developed a double Mitsunobu reaction that allows us a fast implementation of the linker to different acyclic templates. Using the 17-membered macrocycles **9d** and **9f** we were able to adopt the bioactive conformation of gefitinib based on our co-crystal structure of **9d** with EGFR WT. Moreover, both macrocycles demonstrated excellent cellular efficacy in our Ba/F3 cell model, with **9d** showing even superior on-target activity compared to gefitinib. As expected, all quinazoline-based macrocycles were inactive against the gatekeeper mutant T797M just as the two first generation TKIs gefitinib or erlotinib. However, the shorter 14-membered macrocycle **3f** was inactive on the EGFR WT, while it was still highly potent against the most common EGFR Del19 mutation (IC₅₀: 119 nM). In addition, **3f** was still highly effective in the cellular system on both most common EGFR mutations Del19 and L858R (IC₅₀: 198 respectively 386 nM) and the corresponding double mutations Del19/C797S and L858R/C797S (IC₅₀: 148 respectively 750 nM), with outstanding kinome wide selectivity.

According to the latest meta-analysis of clinical data, gefitinib and erlotinib remain the first-line options for the treatment of patients with EGFR mutation in NSCLC and stable brain metastases.³⁵ However, the most common adverse events by the treatment with gefitinib are diarrhea, dry skin or acneiform rash. These adverse effects are related to the mechanism of action by EGFR inhibitors and were not caused by a drug-related allergic reaction.^{36,37} Consistently, inhibitors that are effective on EGFR mutations while ineffective on the EGFR WT could be represent a treatment regimen for such patients, potentially cause less side effects as they are sparing wild type EGFR. In our macrocyclic case study on a quinazoline scaffold, we identified such compound in **3f** with a favourable pharmacological profile. Even though **3f** displayed no activity on the gatekeeper mutation, presumably due to the quinazoline moiety, we believe that is a valuable tool compound for the most common EGFR mutations Del19 and L858R and even the double mutations Del19/C797S and L858R/C797S.

Experimental Section

Differential Scanning Fluorimetry Assay.

Recombinant protein kinase domains with a concentration of 2 μ M were mixed with a 10 μ M compound solution in DMSO, 20 mM HEPES, pH 7.5, and 500 mM NaCl. SYPRO Orange (5000 $\times$, Invitrogen) was added as a fluorescence probe (1 μ L per mL). Subsequently, temperature-dependent protein unfolding profiles were measured, using the QuantStudio™ 5 realtime PCR machine (Thermo Fisher). Excitation and emission filters were set to 465 nm and 590 nm. The temperature was raised with a step rate of 3°C per minute. Data points were analysed with the internal software (Thermal Shift Software™ Version 1.4, Thermo Fisher) using the Boltzmann equation to determine the inflection point of the transition curve. Differences in melting temperature are given as ΔT_m values in °C. Measurements were performed in duplicates.

Kinome-Wide Selectivity Profile.

Compound **3f** was tested at a concentration of 1 μ M against a panel of 468 kinases in the KINOMEScan assay performed by Eurofins Scientific.

Kinase inhibition assays.

Inhibition assays were performed using the HTRF KinEASE tyrosine kinase assay kit (Cisbio). Inhibitors (10 mM DMSO stocks) were dispensed using a D300e dispenser (Hewlett-Packard) and normalized to 1% DMSO. Assay kit buffer containing purified protein at a final concentration of 0.1 nM for EGFR L858R, 0.02 nM for EGFR L858R/T790M, or 10 nM for EGFR WT were dispensed using a Multidrop Combi dispenser (ThermoFisher) and preincubated with the inhibitors at room temperature for 30 min. Reactions were initiated with 100 μ M ATP and allowed to proceed for 30 min at room temperature before being quenched using the detection reagent from the KinEASE assay kit. The FRET signal ratio was measured at 665 and 620 nm using a PHERAstar microplate reader (BMG LABTECH). Data were processed fit to a three-parameter dose-response model with a Hill slope constrained to -1 in GraphPad Prism. The assay was performed three times in triplicate.

Cell Viability Assay.

Ba/F3 cells were plated in 384-well plates and were treated with varying doses of inhibitors in triplicate. EGFR wild type Ba/F3 cells were additionally supplemented with 10 ng/mL EGF. Cell viability was assessed 72 hours after treatment using Cell Titer Glo reagent (Promega) per manufacturer's instructions. Cell viability assays were performed a total of 3 independent experiments.

Western Blot.

Ba/F3 cells expressing various EGFR mutants were treated with varying doses of inhibitors for 8 hours. Cells were then harvested and lysed in RIPA buffer. Protein quantification was performed using Pierce BCA protein assay (Thermo Fisher). 10 μ g protein lysate was resolved on NuPAGE 4-12% Bis-Tris gels (Thermo Fisher), followed by transfer onto

PVDF membranes (Millipore). Membranes were blocked with 3% BSA for 30 minutes on a rocking platform at room temperature. Membranes were subsequently incubated overnight at 4°C with primary antibodies for phospho-EGFR (Cell Signaling #3777), total EGFR (Cell Signaling #4267), phospho-AKT (Cell Signaling #4060), total AKT (Cell Signaling #9272), phospho-ERK1/2 (Cell Signaling #4377), total ERK1/2 (Cell Signaling #9102), and HSP90 (Cell Signaling #4877). The following morning, membranes were incubated with anti-rabbit secondary antibody (Cell Signaling #7074) and imaged on an Amersham Imager 600 chemiluminescence imager using SuperSignal West Dura ECL substrate (Thermo Fisher).

Protein Expression and Purification.

The EGFR kinase domain, spanning residues 696-1022, was cloned into pTriEx with a TEV-cleavable, N-terminal 6xHis-glutathione S-transferase (GST) fusion tag. Mutations were introduced via mismatch PCR and the resulting sequence verified via Sanger sequencing. Recombinant baculovirus was prepared and used to infect Sf9 insect cells. Infected cells were pelleted and resuspended in lysis buffer composed of 50 mM Tris pH 8.0, 500 mM NaCl, 1 mM tris(2-carboxyethyl)phosphine (TCEP), and 5% glycerol, lysed via sonication, and ultracentrifuged at >200,000 g for 1 h. Imidazole pH 8.0 was added to the supernatant for a final concentration of 40 mM and flowed through a column containing Ni-NTA agarose beads. The resin was washed with lysis buffer supplemented with 40 mM imidazole and eluted with lysis buffer containing 200 mM imidazole. Eluted protein was dialyzed overnight in the presence of TEV protease against dialysis buffer containing 50 mM Tris pH 8.0, 500 mM NaCl, 1 mM TCEP, and 5% glycerol prior to being passed through Ni-NTA resin to remove the 6xHis-GST fusion protein and TEV. EGFR was lastly purified via size exclusion chromatography in 50 mM Tris pH 8.0, 500 mM NaCl, 1 mM TCEP, and 5% glycerol.

Tissue culture.

BaF/3 cells were a generous gift from the laboratory of Dr. David Weinstock (in 2014). Ba/F3 cells expressing various EGFR mutants were previously generated and characterized.³⁸⁻⁴¹ Cells were cultured in RPMI-1640 media supplemented with 10% fetal bovine serum in addition to 1% penicillin and streptomycin. Ba/F3 cells expressing wildtype EGFR were additionally supplemented with 10 ng/mL EGF (Life Technologies). Cells were maintained in a 37°C humidified incubator with 5% CO₂.

Crystallization and structure determination.

Purified EGFR T790M/V948R and wild type EGFR kinase domain was crystallized at approximately 3 mg/mL with 10 mM MgCl₂ and 1 mM adenylyl-imidodiphosphate (AMP-PNP) via hanging drop vapor diffusion at room temperature. EGFR T790M/V948R crystallized in 0.1 M Bis-Tris pH 5.0-6.0 and 20-30% (w/v) PEG 3,350, and wild type EGFR crystallized in 0.1 M MES pH 6.5 and 0.8 M sodium citrate. Crystals were soaked with 0.5 mM inhibitor overnight. EGFR T790M/V948R and wild type EGFR crystals were briefly cryoprotected in well solution containing 20% or 30% ethylene glycol, respectively, and flash frozen in liquid nitrogen.

Diffraction data were collected at the Advanced Photon Source at the Argonne National Laboratory on NE-CAT beamline 24-ID-C at 100 K. Data were indexed, integrated, and scaled using Dials via xia2 compiled by SGrid. Structures were phased via molecular replacement with PDB 5D41 or 2GS2. Refinement was performed using Phenix with iterative rounds of manual model building in Coot. Structures have been deposited in the Protein Data Bank (PDB) with the accession codes 7U98, 7U99, and 7U9A. Data collection and the refinement statistics are summarized in Table S5.

Stability assay.

The compound (25 μ L, concentration 500 μ M in DMSO) was diluted with 225 μ L of either a 0.1 M HCl or PBS buffer to get a final concentration of 50 μ M. The solution was quantified by HPLC every 23 minutes to cover a total time of 184 minutes (mobile phase: 0.1% TFA in water (A) and 0.1% TFA in acetonitrile (B)). The following gradient was used: 0 min. 2% B - 2 min. 2% B - 10 min. 98% B - 15 min. 98% B - 17 min. 2% B - 21 min. 2% B (flow rate of 1 mL/min). The amount was determined by an external calibration curve and data was obtained in three independent experiments.

Solubility assay.

1 – 5 mg of the compound were weight into a Whatman UNIPREP filter and 2 mL of the desired solvent was added. The solution had to be saturated while the compound was incubated for 24 h at 37 °C on a shaking plate. The solution was filtered and quantified by HPLC using 0.1% TFA in water (A) and 0.1% TFA in acetonitrile (B) as a mobile phase. The following gradient was used: 0 min. 2% B - 2 min. 2% B - 10 min. 98% B - 15 min. 98% B - 17 min. 2% B - 21 min. 2% B (flow rate of 1 mL/min). The amount was determined by an external calibration curve and data was obtained in technical triplicates.

Chemistry.

The synthesis of compounds will be explained in the following and the analytical data for them can be found in the Supporting Information. All commercial chemicals were purchased from common suppliers with a purity $\geq$ 95% and were used without further purification. The solvents with an analytical grade were obtained from VWR Chemicals and Merck and all dry solvents from Acros Organics. All reactions were proceeded under an argon atmosphere. The thin layer chromatography was done with silica gel on aluminum foils (60 Å pore diameter) obtained from Macherey-Nagel and visualized with ultraviolet light (λ = 254 and 365 nm). The purification of the compounds was done by flash chromatography. A puriFlash XS 420 device with a UV-VIS multiwave detector (200–400 nm) from Interchim was used with pre-packed normal-phase PF-SIHP silica columns with particle sizes of 15 and 30 μ m (Interchim). Preparative purification by HPLC was carried out on an Agilent 1260 Infinity II device using an Eclipse XDB-C18 (Agilent, 21.2 x 250mm, 7 μ m) reversed phase column. A suitable gradient (flow rate 21 mL/min.) was used, with 0.1% TFA in water (A) and 0.1% TFA in acetonitrile (B), as a mobile phase. The nuclear magnetic resonance spectroscopy (NMR) was performed with DPX250, AV300, AV400 or AV500 MHz spectrometers from Bruker. Chemical shifts (δ) are reported in parts per million (ppm). DMSO-d₆, chloroform-d and methylene chloride-d₂ was used as a solvent, and the spectra were calibrated to the solvent signal: 2.50 ppm (1H NMR) or 39.52 ppm (13C NMR) for DMSO-d₆, 7.26 ppm

(¹H NMR) or 77.16 ppm (¹³C NMR) for chloroform-d and 5.32 ppm (¹H NMR) or 54.00 ppm (¹³C NMR) for methylene chloride-d₂. Coupling constants (J) were reported in hertz (Hz) and multiplicities were designated as followed: s (singlet), d (doublet), dd (doublet of doublet), t (triplet), dt (doublet of triplets), td (triplet of doublets), ddd (doublet of doublet of doublet), q (quartet), m (multiplet). Mass spectra were measured on a Surveyor MSQ device from ThermoFisher measuring in the positive- or negative-ion mode. Final compounds were additionally characterized by HRMS using a MALDI LTQ Orbitrap XL from ThermoScientific. The purity of the final compounds was determined by HPLC using an Agilent 1260 Infinity II device with a 1260 DAD HS detector (G7117C; 254 nm, 280 nm, 310 nm) and a LC/MSD device (G6125B, ESI pos. 100-1000). The compounds were analyzed on a Poroshell 120 EC-C18 (Agilent, 3 x 150 mm, 2.7 μm) reversed phase column using 0.1% formic acid in water (A) and 0.1% formic acid in acetonitrile (B) as a mobile phase. The following gradient was used: 0 min 5% B - 2 min 5% B - 8 min 98% B - 10 min 98% B (flow rate of 0.5 mL/min). UV-detection was performed at 254, 280 and 310 nm and all compounds used for further biological characterizations showed a purity ≥95%.

4-((2-hydroxyphenyl)amino)-7-methoxyquinazolin-6-ol (2a).

The title compound was prepared according to the general procedure 1 using 2-aminophenol (57 mg, 0.52 mmol) to obtain the product (80 mg, 60%) as a white solid. ¹H NMR (400 MHz, DMSO-d₆) δ 10.75 (s, 1H), 10.47 (s, 1H), 9.88 (s, 1H), 8.67 (s, 1H), 7.95 (s, 1H), 7.39 (d, J = 2.1 Hz, 1H), 7.32 (dd, J = 7.9, 1.6 Hz, 1H), 7.26 – 7.17 (m, 1H), 7.03 (dd, J = 8.2, 1.5 Hz, 1H), 6.89 (t, J = 7.6 Hz, 1H), 4.01 (s, 3H). ¹³C NMR (75 MHz, DMSO) δ 155.88, 152.55, 148.33, 128.45, 128.18, 123.90, 118.97, 116.52, 107.44, 107.05, 56.42. MS-ESI *m/z* [M + H]⁺: calcd 284.3, found 284.2. HRMS *m/z* [M + H]⁺: calcd 284.1030, found 284.1035. HPLC: *t_R* = 6.27, purity ≥ 95% (UV: 254/ 280 nm).

4-((3-chloro-2-hydroxyphenyl)amino)-7-methoxyquinazolin-6-ol (2b).

The title compound was prepared according to the general procedure 1 using 2-amino-6-chlorophenol (75 mg, 0.52 mmol) to obtain the product (137 mg, 91%) as a beige solid. ¹H NMR (400 MHz, DMSO-d₆) δ 10.94 (s, 1H), 10.52 (s, 1H), 9.90 (s, 1H), 8.70 (s, 1H), 7.97 (s, 1H), 7.40 (dt, J = 6.8, 1.5 Hz, 2H), 7.34 – 7.22 (m, 1H), 6.93 (td, J = 8.0, 1.5 Hz, 1H), 4.02 (s, 3H). ¹³C NMR (101 MHz, DMSO) δ 156.01, 149.03, 148.32, 148.10, 128.95, 127.42, 125.97, 121.52, 119.71, 107.85, 107.39, 99.73, 56.45. MS-ESI *m/z* [M + H]⁺: calcd 318.7, found 318.1. HRMS *m/z* [M + H]⁺: calcd 318.0640, found 318.0648. HPLC: *t_R* = 6.63, purity ≥ 95% (UV: 254/ 280 nm).

4-((4-chloro-2-hydroxyphenyl)amino)-7-methoxyquinazolin-6-ol (2c).

The title compound was prepared according to the general procedure 1 using 2-amino-5-chlorophenol (75 mg, 0.52 mmol) to obtain the product (100 mg, 66%) as a beige solid. ¹H NMR (400 MHz, DMSO-d₆) δ 10.78 (s, 1H), 10.52 (s, 2H), 8.69 (s, 1H), 7.93 (s, 1H), 7.56 – 7.26 (m, 2H), 7.09 (d, J = 2.3 Hz, 1H), 6.95 (dd, J = 8.4, 2.3 Hz, 1H), 4.01 (s, 3H). ¹³C NMR (101 MHz, DMSO) δ 156.04, 153.66, 148.43, 132.03, 129.58, 123.19, 118.82, 116.21, 107.43, 106.96, 56.44. MS-ESI *m/z* [M + H]⁺: calcd 318.7, found 318.1. HRMS *m/z* [M + H]⁺: calcd 318.0640, found 318.0647. HPLC: *t_R* = 6.74, purity ≥ 95% (UV: 254/ 280 nm).

4-((5-chloro-2-hydroxyphenyl)amino)-7-methoxyquinazolin-6-ol (2d).

The title compound was prepared according to the general procedure 1 using 2-amino-4-chlorophenol (75 mg, 0.52 mmol) to obtain the product (128 mg, 85%) as a beige solid. ¹H NMR (400 MHz, DMSO-d₆) δ 10.78 (s, 1H), 10.54 (s, 1H), 10.26 (s, 1H), 8.73 (s, 1H), 7.93 (s, 1H), 7.47 (d, J = 2.6 Hz, 1H), 7.38 (d, J = 3.0 Hz, 1H), 7.25 (dd, J = 8.7, 2.6 Hz, 1H), 7.05 (d, J = 8.7 Hz, 1H), 4.02 (s, 3H). ¹³C NMR (101 MHz, DMSO) δ 156.08, 152.36, 151.59, 148.48, 128.01, 127.68, 121.84, 117.77, 107.49, 106.91, 56.45. MS-ESI *m/z* [M + H]⁺: calcd 318.7, found 318.1. HRMS *m/z* [M + H]⁺: calcd 318.0640, found 318.0644. HPLC: t_R = 6.76, purity ≥ 95% (UV: 254/ 280 nm).

4-((2-chloro-6-hydroxyphenyl)amino)-7-methoxyquinazolin-6-ol (2e).

The title compound was prepared according to the general procedure 2 using 2-amino-3-chlorophenol (75 mg, 0.52 mmol) to obtain the product (79 mg, 52%) as a brown solid. ¹H NMR (400 MHz, DMSO-d₆) δ 9.76 (s, 1H), 9.55 (s, 1H), 8.99 (s, 1H), 8.19 (s, 1H), 7.71 (s, 1H), 7.28 – 7.08 (m, 2H), 6.97 (d, J = 8.0 Hz, 1H), 6.89 (d, J = 8.2 Hz, 1H), 3.96 (s, 3H). ¹³C NMR (101 MHz, DMSO) δ 157.92, 155.72, 153.68, 152.61, 146.20, 145.88, 133.53, 128.01, 124.32, 119.56, 114.98, 109.34, 106.91, 105.90, 55.81. MS-ESI *m/z* [M + H]⁺: calcd 318.7, found 318.1. HRMS *m/z* [M + H]⁺: calcd 318.0640, found 318.0640. HPLC: t_R = 6.51, purity ≥ 95% (UV: 254/ 280 nm).

4-((5-chloro-4-fluoro-2-hydroxyphenyl)amino)-7-methoxyquinazolin-6-ol (2f).

The title compound was prepared according to the general procedure 1 using 2-amino-4-chloro-5-fluorophenol (127 mg, 0.78 mmol) to obtain the product (239 mg, 100%) as a beige solid. ¹H NMR (250 MHz, DMSO-d₆) δ 10.79 (s, 2H), 10.55 (s, 1H), 8.73 (s, 1H), 7.92 (s, 1H), 7.60 (d, J = 8.2 Hz, 1H), 7.38 (s, 1H), 7.06 (dd, J = 10.8, 3.6 Hz, 1H), 4.02 (s, 3H). ¹³C NMR (101 MHz, DMSO) δ 159.23, 156.32 (d, J = 246.3 Hz), 156.11, 153.27 (d, J = 10.5 Hz), 148.48, 148.09, 134.26, 129.41, 121.48 (d, J = 2.9 Hz), 108.12 (d, J = 18.9 Hz), 107.45, 106.95, 104.67 (d, J = 24.0 Hz), 99.79, 56.46. MS-ESI *m/z* [M + H]⁺: calcd 335.7, found 336.0. HRMS *m/z* [M + H]⁺: calcd 335.0546, found 335.0546. HPLC: t_R = 6.97, purity ≥ 95% (UV: 254/ 280 nm).

1⁷-methoxy-4,7,10-trioxa-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacyclodecaphane (3a).

The title compound was prepared according to the general procedure 3 using **2a** (80 mg, 0.28 mmol) to obtain the product (17 mg, 17%) as a white solid. ¹H NMR (300 MHz, DMSO-d₆) δ 8.87 (s, 1H), 8.61 (s, 1H), 8.50 (dd, J = 8.0, 1.7 Hz, 1H), 8.44 (s, 1H), 7.25 (dd, J = 7.9, 1.5 Hz, 1H), 7.19 (s, 1H), 7.12 (td, J = 7.8, 1.5 Hz, 1H), 7.02 (td, J = 7.7, 1.7 Hz, 1H), 4.59 – 4.53 (m, 2H), 4.26 – 4.20 (m, 2H), 3.98 – 3.92 (m, 5H), 3.90 – 3.85 (m, 2H). ¹³C NMR (75 MHz, DMSO) δ 155.79, 154.68, 153.30, 148.40, 147.77, 146.30, 131.68, 123.24, 122.41, 118.32, 117.16, 109.38, 106.96, 104.90, 71.92, 71.49, 70.67, 69.36, 55.92. MS-ESI *m/z* [M + H]⁺: calcd 354.4, found 354.2. HRMS *m/z* [M + H]⁺: calcd 354.1448, found 354.1453. HPLC: t_R = 7.36, purity ≥ 95% (UV: 254/ 280 nm).

3³-chloro-1⁷-methoxy-4,7,10-trioxa-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacyclodecaphane (3b).

The title compound was prepared according to the general procedure 3 using **2b** (120 mg, 0.38 mmol) to obtain the product (20 mg, 14%) as a white solid. ¹H NMR (250 MHz, DMSO-d₆) δ 8.77 – 8.69 (m, 2H), 8.65 – 8.60 (m, 2H), 7.30 – 7.09 (m, 3H), 4.64 – 4.57 (m, 2H), 4.27 – 4.20 (m, 2H), 4.00 – 3.92 (m, 5H), 3.92 – 3.87 (m, 2H). ¹³C NMR (75 MHz, DMSO) δ 155.81, 155.13, 153.54, 148.69, 146.69, 143.55, 135.17, 126.90, 126.27, 123.31, 118.46, 109.37, 107.38, 105.56, 73.47, 72.18, 70.86, 69.67, 56.47. MS-ESI *m/z* [M + H]⁺: calcd 388.8, found 388.2. HRMS *m/z* [M + H]⁺: calcd 388.1059, found 388.1058. HPLC: t_R = 7.97, purity ≥ 95% (UV: 254/ 280 nm).

3⁴-chloro-1⁷-methoxy-4,7,10-trioxa-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacyclodecaphane (3c).

The title compound was prepared according to the general procedure 3 using **2c** (100 mg, 0.31 mmol) to obtain the product (10 mg, 8%) as a white solid. ¹H NMR (300 MHz, DMSO-d₆) δ 8.86 (s, 1H), 8.63 (s, 1H), 8.46 (d, J = 8.7 Hz, 1H), 8.43 (s, 1H), 7.41 (d, J = 2.4 Hz, 1H), 7.24 – 7.16 (m, 2H), 4.61 – 4.55 (m, 2H), 4.34 – 4.27 (m, 2H), 3.98 – 3.92 (m, 5H), 3.91 – 3.86 (m, 2H). ¹³C NMR (75 MHz, DMSO) δ 155.90, 154.92, 153.14, 148.67, 148.50, 130.69, 125.90, 122.90, 119.40, 117.32, 109.45, 106.78, 105.05, 72.02, 71.52, 70.57, 69.20, 55.99. MS-ESI *m/z* [M + H]⁺: calcd 388.8, found 388.2. HRMS *m/z* [M + H]⁺: calcd 388.1059, found 388.1059. HPLC: t_R = 7.81, purity ≥ 95% (UV: 254/ 280 nm).

3⁵-chloro-1⁷-methoxy-4,7,10-trioxa-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacyclodecaphane (3d).

The title compound was prepared according to the general procedure 3 using **2d** (110 mg, 0.35 mmol) to obtain the product (15 mg, 11%) as a white solid. ¹H NMR (300 MHz, DMSO-d₆) δ 8.88 (s, 1H), 8.67 (s, 1H), 8.56 (d, J = 2.6 Hz, 1H), 8.45 (s, 1H), 7.29 (d, J = 8.6 Hz, 1H), 7.22 (s, 1H), 7.07 (dd, J = 8.6, 2.6 Hz, 1H), 4.63 – 4.54 (m, 2H), 4.30 – 4.19 (m, 2H), 4.00 – 3.92 (m, 5H), 3.92 – 3.85 (m, 2H). ¹³C NMR (75 MHz, DMSO) δ 155.55, 154.89, 153.16, 148.54, 146.56, 146.46, 132.93, 126.96, 121.68, 118.70, 117.54, 109.39, 106.97, 104.89, 72.39, 71.46, 70.73, 69.24, 55.98. MS-ESI *m/z* [M + H]⁺: calcd 388.8, found 388.2. HRMS *m/z* [M + H]⁺: calcd 388.1059, found 388.1059. HPLC: t_R = 7.94, purity ≥ 95% (UV: 254/ 280 nm).

3⁶-chloro-1⁷-methoxy-4,7,10-trioxa-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacyclodecaphane (3e).

The title compound was prepared according to the general procedure 3 using **2e** (55 mg, 0.17 mmol) to obtain the product (3 mg, 4%) as a white solid. ¹H NMR (500 MHz, DMSO-d₆) δ 9.22 (s, 1H), 8.27 (s, 1H), 7.82 (s, 1H), 7.35 (d, J = 8.6 Hz, 1H), 7.22 – 7.06 (m, 3H), 4.19 – 3.74 (m, 9H), 3.53 – 3.45 (m, 2H). ¹³C NMR (126 MHz, DMSO) δ 158.19, 156.21, 153.88, 153.10, 147.84, 146.35, 133.74, 121.93, 121.30, 120.93, 111.48, 108.50, 106.96, 102.30, 69.41, 69.30, 68.86, 68.54, 55.72. MS-ESI *m/z* [M + H]⁺: calcd found 388.3. HRMS *m/z* [M + H]⁺: calcd 388.1059, found 388.1054. HPLC: t_R = 7.37, purity ≥ 95% (UV: 254/ 280 nm).

3⁵-chloro-3⁴-fluoro-17-methoxy-4,7,10-trioxa-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacyclodecaphane (3f).

The title compound was prepared according to the general procedure 3 using **2f** (150 mg, 0.45 mmol) to obtain the product (33 mg, 18%) as a white solid. ¹H NMR (250 MHz, DMSO-d₆) δ 8.77 (s, 1H), 8.67 (s, 1H), 8.57 (d, J = 8.1 Hz, 1H), 8.41 (s, 1H), 7.50 (d, J = 10.2 Hz, 1H), 7.23 (s, 1H), 4.67 – 4.50 (m, 2H), 4.38 – 4.23 (m, 2H), 3.98 – 3.92 (m, 5H), 3.92 – 3.85 (m, 2H). ¹³C NMR (126 MHz, DMSO) δ 158.73, 155.98, 154.96, 153.09, 148.47, 129.18, 118.95, 112.23, 109.49, 106.89, 106.64, 106.44, 105.14, 105.12, 72.13, 71.54, 70.41, 69.06, 55.99. MS-ESI *m/z* [M + H]⁺: calcd 405.8, found 406.0. HRMS *m/z* [M + H]⁺: calcd 406.0964, found 406.0946. HPLC: t_R = 8.28, purity ≥ 95% (UV: 254/280 nm).

4-chloro-6-((6-chlorohexyl)oxy)-7-methoxyquinazoline (4).

The title compound was prepared according to the general procedure 4 using 6-chloro-1-hexanol (324 mg, 2.37 mmol) and heated for 3 h to obtain the product (487 mg, 78%) as a white solid with impurities. ¹H NMR (300 MHz, DMSO-d₆) δ 8.86 (s, 1H), 7.43 (s, 1H), 7.36 (s, 1H), 4.18 (t, J = 6.5 Hz, 2H), 4.01, (s, 3H), 3.65 (t, J = 6.6 Hz, 2H), 1.90 – 1.69 (m, 4H), 1.53 – 1.43 (m, 4H). MS-ESI *m/z* [M + H]⁺: calcd 330.2, found 329.1.

4-chloro-6-(2-(2-(2-chloroethoxy)ethoxy)ethoxy)-7-methoxyquinazoline (5).

The title compound was prepared according to the general procedure 4 using 2-[2-(2-chloroethoxy)ethoxy]ethanol (400 mg, 2.37 mmol) and heated for 5 h to obtain the product (520 mg, 76%) as a yellow solid with impurities. MS-ESI *m/z* [M + Na]⁺: calcd 383.1, found 383.2.

2-((6-((6-chlorohexyl)oxy)-7-methoxyquinazolin-4-yl)amino)phenol (6a).

The title compound was prepared according to the general procedure 5 using **5** (124 mg, 0.38 mmol) and 2-aminophenol (41 mg, 0.38 mmol) to obtain the product (105 mg, 70%) as a brown oil. ¹H NMR (300 MHz, DMSO-d₆) δ 10.81 (s, 1H), 8.65 (s, 1H), 8.17 (s, 1H), 7.38 – 7.28 (m, 2H), (t, J = 7.7 Hz, 1H), 7.03 (d, J = 8.0 Hz, 1H), 6.89 (t, J = 7.5 Hz, 1H), 4.17 (t, J = 6.5 Hz, 2H), 3.98 (s, 3H), 3.65 (t, J = 6.5 Hz, 2H), 1.91 – 1.69 (m, 4H), 1.57 – 1.42 (m, 4H). ¹³C NMR (75 MHz, DMSO) δ 158.83, 155.96, 152.44, 149.21, 149.06, 136.70, 128.27, 128.15, 124.07, 119.00, 116.69, 107.17, 104.28, 100.82, 69.12, 56.34, 45.35, 31.95, 28.26, 26.00, 24.80. MS-ESI *m/z* [M + H]⁺: calcd 402.9, found 402.1. HRMS *m/z* [M + H]⁺: calcd 402.1579, found 402.1693. HPLC: t_R = 8.00, purity ≥ 95% (UV: 254/280 nm).

2-chloro-6-((6-((6-chlorohexyl)oxy)-7-methoxyquinazolin-4-yl)amino)phenol (6b).

The title compound was prepared according to the general procedure 6 using **5** (124 mg, 0.38 mmol) and 2-amino-6-chlorophenol (54 mg, 0.38 mmol) to obtain the product (107 mg, 65%) as a beige solid. ¹H NMR (300 MHz, DMSO-d₆) δ 11.30 (s, 1H), 9.90 (s, 1H), 8.75 (s, 1H), 8.28 (s, 1H), 7.48 – 7.37 (m, 2H), 7.30 (dd, J = 7.9, 1.6 Hz, 1H), 6.95 (t, J = 8.0 Hz, 1H), 4.21 (t, J = 6.5 Hz, 2H), 4.00 (s, 3H), 3.65 (t, J = 6.6 Hz, 2H), 1.97 – 1.69 (m, 4H), 1.56 – 1.42 (m, 4H). ¹³C NMR (75 MHz, DMSO) δ 159.63, 156.38, 149.41, 148.92,

148.62, 129.00, 127.38, 125.86, 121.69, 119.79, 107.36, 104.88, 99.66, 69.30, 56.48, 45.36, 31.95, 28.22, 26.00, 24.80. MS-ESI m/z $[M + H]^+$: calcd 437.3, found 437.1. HRMS m/z $[M + H]^+$: calcd 436.1189, found 436.1187. HPLC: t_R = 8.34, purity $\geq$ 95% (UV: 254/ 280 nm).

5-chloro-2-((6-((6-chlorohexyl)oxy)-7-methoxyquinazolin-4-yl)amino)phenol (6c).

The title compound was prepared according to the general procedure 5 using **5** (132 mg, 0.40 mmol) and 2-amino-5-chlorophenol (58 mg, 0.40 mmol) to obtain the product (156 mg, 89%) as a beige solid. 1H NMR (300 MHz, DMSO- d_6) δ 11.00 (s, 1H), 10.52 (s, 1H), 8.71 (s, 1H), 8.19 (s, 1H), 7.41 – 7.32 (m, 2H), 7.11 (d, J = 2.2 Hz, 1H), 6.96 (dd, J = 8.4, 2.3 Hz, 1H), 4.18 (t, J = 6.5 Hz, 2H), 3.99 (s, 3H), 3.65 (t, J = 6.5 Hz, 2H), 1.90 – 1.69 (m, 4H), 1.61 – 1.42 (m, 4H). ^{13}C NMR (75 MHz, DMSO) δ 159.08, 156.27, 153.64, 149.41, 148.64, 131.99, 129.63, 123.10, 118.87, 116.32, 107.02, 104.36, 100.08, 69.20, 56.43, 45.34, 31.94, 28.23, 25.99, 24.79. MS-ESI m/z $[M + H]^+$: calcd 437.3, found 436.3. HRMS m/z $[M + H]^+$: calcd 436.1189, found 436.1190. HPLC: t_R = 8.28, purity $\geq$ 95% (UV: 254/ 280 nm).

4-chloro-2-((6-((6-chlorohexyl)oxy)-7-methoxyquinazolin-4-yl)amino)phenol (6d).

The title compound was prepared according to the general procedure 5 using **5** (132 mg, 0.40 mmol) and 2-amino-4-chlorophenol (58 mg, 0.40 mmol) to obtain the product (172 mg, 98%) as a yellow solid. 1H NMR (300 MHz, DMSO- d_6) δ 11.01 (s, 1H), 10.30 (s, 1H), 8.73 (s, 1H), 8.20 (s, 1H), 7.45 (d, J = 2.6 Hz, 1H), 7.37 (s, 1H), 7.25 (dd, J = 8.7, 2.7 Hz, 1H), 7.06 (d, J = 8.7 Hz, 1H), 4.18 (t, J = 6.5 Hz, 2H), 3.99 (s, 3H), 3.65 (t, J = 6.6 Hz, 2H), 1.91 – 1.73 (m, 4H), 1.57 – 1.41 (m, 4H). ^{13}C NMR (75 MHz, DMSO) δ 158.96, 156.26, 151.58, 149.40, 148.84, 136.04, 127.97, 127.79, 125.19, 121.87, 117.93, 107.13, 104.35, 100.27, 69.21, 56.42, 45.35, 31.95, 28.25, 26.00, 24.81. MS-ESI m/z $[M + H]^+$: calcd 437.3, found 436.3. HRMS m/z $[M + H]^+$: calcd 436.1189, found 436.1188. HPLC: t_R = 8.35, purity $\geq$ 95% (UV: 254/ 280 nm).

3-chloro-2-((6-((6-chlorohexyl)oxy)-7-methoxyquinazolin-4-yl)amino)phenol (6e).

The title compound was prepared according to the general procedure 5 using **5** (132 mg, 0.40 mmol) and 2-amino-3-chlorophenol (58 mg, 0.40 mmol) to obtain the product (87 mg, 50%) as a brown solid. 1H NMR (300 MHz, DMSO- d_6) δ 11.06 (s, 1H), 10.41 (s, 1H), 8.73 (s, 1H), 8.22 (s, 1H), 7.43 (s, 1H), 7.26 (t, J = 8.1 Hz, 1H), 7.06 (s, 1H), 7.04 (s, 1H), 4.19 (t, J = 6.5 Hz, 2H), 4.00 (s, 3H), 3.65 (t, J = 6.5 Hz, 2H), 1.87 – 1.72 (m, 4H), 1.57 – 1.42 (m, 4H). ^{13}C NMR (75 MHz, DMSO) δ 159.67, 156.48, 155.19, 149.59, 148.76, 135.43, 132.44, 129.50, 121.75, 119.62, 115.33, 106.73, 104.22, 99.91, 69.22, 56.49, 45.33, 31.94, 28.22, 25.98, 24.76. MS-ESI m/z $[M + H]^+$: calcd 437.3, found 436.3. HRMS m/z $[M + H]^+$: calcd 436.1189, found 436.1187. HPLC: t_R = 8.14, purity $\geq$ 95% (UV: 254/ 280 nm).

4-chloro-2-((6-((6-chlorohexyl)oxy)-7-methoxyquinazolin-4-yl)amino)-5-fluorophenol (6f).

The title compound was prepared according to the general procedure 5 using **5** (123 mg, 0.37 mmol) and 2-amino-4-chloro-5-fluorophenol (60 mg, 0.37 mmol) to obtain the product (155 mg, 92%) as a brown solid. 1H NMR (300 MHz, DMSO- d_6) δ 11.02 (s, 2H), 8.72 (s, 1H), 8.20 (s, 1H), 7.57 (d, J = 8.0 Hz, 1H), 7.36 (s, 1H), 7.07 (d, J = 10.8 Hz, 1H), 4.17 (t, J = 6.3 Hz, 2H), 3.98 (s, 3H), 3.65 (t, J = 6.4 Hz, 2H), 1.92 – 1.69 (m, 4H), 1.57 – 1.40

(m, 4H). ¹³C NMR (75 MHz, DMSO) δ 159.13, 156.19, 156.18 (d, J = 246.0 Hz), 153.13 (d, J = 10.5 Hz), 149.33, 148.82, 136.05, 129.33, 121.45 (d, J = 2.9 Hz), 108.10 (d, J = 19.0 Hz), 107.05, 104.69 (d, J = 24.0 Hz), 104.31, 100.24, 69.21, 56.38, 45.32, 31.93, 28.22, 25.98, 24.77. MS-ESI *m/z* [M + H]⁺: calcd 455.3, found 454.3. HRMS *m/z* [M + H]⁺: calcd 454.1095, found 454.1094. HPLC: *t_R* = 8.42, purity ≥ 95% (UV: 254/ 280 nm).

2-((6-(2-(2-(2-chloroethoxy)ethoxy)ethoxy)-7-methoxyquinazolin-4-yl)amino)phenol (7a).

The title compound was prepared according to the general procedure 6 using **5** (200 mg, 0.55 mmol) and 2-aminophenol (60 mg, 0.55 mmol) to obtain the product (105 mg, 44%) as a yellow solid. ¹H NMR (250 MHz, DMSO-*d*₆) δ 11.09 (s, 1H), 9.90 (s, 1H), 8.71 (s, 1H), 8.28 (s, 1H), 7.44 – 7.26 (m, 2H), 7.22 (t, J = 7.7 Hz, 1H), 7.04 (d, J = 7.9 Hz, 1H), 6.90 (t, J = 7.5 Hz, 1H), 4.32 (t, J = 4.7 Hz, 2H), 3.99 (s, 3H), 3.88 (t, J = 4.6 Hz, 2H), 3.75 – 3.56 (m, 8H). ¹³C NMR (75 MHz, DMSO) δ 156.23, 152.56, 149.20, 147.79, 145.57, 128.61, 128.27, 123.63, 119.04, 116.64, 107.67, 106.90, 70.56, 69.95, 69.69, 68.74, 68.45, 56.44, 43.57. MS-ESI *m/z* [M + H]⁺: calcd 434.2, found 434.2. HRMS *m/z* [M + H]⁺: calcd 434.1477, found 434.1476. HPLC: *t_R* = 7.19, purity ≥ 95% (UV: 254/ 280 nm).

2-chloro-6-((6-(2-(2-(2-chloroethoxy)ethoxy)ethoxy)-7-methoxyquinazolin-4-yl)amino)phenol (7b).

The title compound was prepared according to the general procedure 6 using **5** (246 mg, 0.68 mmol) and 2-amino-6-chlorophenol (98 mg, 0.68 mmol) to obtain the product (88 mg, 28%) as a pale green solid. ¹H NMR (300 MHz, DMSO-*d*₆) δ 11.37 (s, 1H), 9.90 (s, 1H), 8.75 (s, 1H), 8.36 (s, 1H), 7.48 – 7.39 (m, 2H), 7.30 (dd, J = 7.9, 1.6 Hz, 1H), 6.94 (t, J = 8.0 Hz, 1H), 4.35 (t, J = 4.5 Hz, 2H), 4.00 (s, 3H), 3.88 (t, J = 4.7 Hz, 2H), 3.75 – 3.59 (m, 8H). ¹³C NMR (75 MHz, DMSO) δ 159.64, 156.31, 149.17, 148.87, 148.68, 135.21, 128.97, 127.33, 125.86, 121.70, 119.79, 107.33, 105.11, 99.71, 70.56, 69.95, 69.67, 68.91, 68.45, 56.47, 43.58. MS-ESI *m/z* [M + H]⁺: calcd 468.1, found 468.1. HRMS *m/z* [M + H]⁺: calcd 468.1088, found 468.1092. HPLC: *t_R* = 7.50, purity ≥ 95% (UV: 254/ 280 nm).

5-chloro-2-((6-(2-(2-(2-chloroethoxy)ethoxy)ethoxy)-7-methoxyquinazolin-4-yl)amino)phenol (7c).

The title compound was prepared according to the general procedure 5 using **5** (246 mg, 0.68 mmol) and 2-amino-5-chlorophenol (98 mg, 0.68 mmol) to obtain the product (130 mg, 41%) as a pale yellow solid. ¹H NMR (300 MHz, DMSO-*d*₆) δ 11.10 (s, 1H), 10.54 (s, 1H), 8.73 (s, 1H), 8.26 (s, 1H), 7.42 – 7.31 (m, 2H), 7.11 (d, J = 2.3 Hz, 1H), 6.95 (dd, J = 8.4, 2.3 Hz, 1H), 4.32 (t, J = 4.5 Hz, 2H), 4.00 (s, 3H), 3.88 (d, J = 4.7 Hz, 2H), 3.77 – 3.58 (m, 8H). ¹³C NMR (75 MHz, DMSO) δ 159.36, 156.28, 153.66, 149.24, 148.79, 132.07, 129.63, 123.02, 118.86, 116.32, 106.94, 104.63, 99.92, 70.56, 69.95, 69.68, 68.80, 68.45, 56.45, 43.57. MS-ESI *m/z* [M + H]⁺: calcd 468.1, found 468.2. HRMS *m/z* [M + H]⁺: calcd 468.1088, found 468.1102. HPLC: *t_R* = 7.54, purity ≥ 95% (UV: 254/ 280 nm).

4-chloro-2-((6-(2-(2-(2-chloroethoxy)ethoxy)ethoxy)-7-methoxyquinazolin-4-yl)amino)phenol (7d).

The title compound was prepared according to the general procedure 6 using **5** (246 mg, 0.68 mmol) and 2-amino-4-chlorophenol (98 mg, 0.68 mmol) to obtain the product (104 mg, 33%) as a yellow solid. ¹H NMR (300 MHz, DMSO-d₆) δ 11.18 (s, 1H), 10.27 (s, 1H), 8.77 (s, 1H), 8.29 (s, 1H), 7.48 – 7.35 (m, 2H), 7.26 (dt, J = 8.8, 2.2 Hz, 1H), 7.11 – 7.02 (m, 1H), 4.32 (t, J = 4.9 Hz, 2H), 4.00 (s, 3H), 3.88 (t, J = 4.6 Hz, 2H), 3.74 – 3.57 (m, 8H). ¹³C NMR (75 MHz, DMSO) δ 159.21, 156.37, 151.68, 149.28, 148.82, 129.03, 128.18, 127.87, 124.96, 121.87, 117.94, 106.98, 104.68, 70.56, 69.95, 69.68, 68.83, 68.45, 56.47, 43.57. MS-ESI *m/z* [M + H]⁺: calcd 468.1, found 468.2. HRMS *m/z* [M + H]⁺: calcd 468.1088, found 468.1105. HPLC: t_R = 7.58, purity ≥ 95% (UV: 254/ 280 nm).

3-chloro-2-((6-(2-(2-(2-chloroethoxy)ethoxy)ethoxy)-7-methoxyquinazolin-4-yl)amino)phenol (7e).

The title compound was prepared according to the general procedure 5 using **5** (200 mg, 0.55 mmol) and 2-amino-3-chlorophenol (82 mg, 0.55 mmol) to obtain the product (119 mg, 46%) as a yellow solid. ¹H NMR (300 MHz, DMSO-d₆) δ 11.12 (s, 1H), 10.41 (s, 1H), 8.76 (s, 1H), 8.27 (s, 1H), 7.45 – 7.39 (m, 1H), 7.27 (t, J = 8.0 Hz, 1H), 7.10 – 7.00 (m, 2H), 4.33 (t, J = 4.3 Hz, 2H), 4.01 (s, 3H), 3.89 (t, J = 4.8 Hz, 2H), 3.74 – 3.58 (m, 8H). ¹³C NMR (75 MHz, DMSO) δ 159.80, 156.52, 155.15, 149.43, 148.80, 135.18, 132.38, 129.58, 121.65, 119.66, 115.35, 106.62, 104.45, 99.83, 70.55, 69.94, 69.68, 68.77, 68.43, 56.52, 43.56. MS-ESI *m/z* [M + H]⁺: calcd 468.1, found 468.2. HRMS *m/z* [M + H]⁺: calcd 468.1088, found 468.1089. HPLC: t_R = 7.36, purity ≥ 95% (UV: 254/ 280 nm).

4-chloro-2-((6-(2-(2-(2-chloroethoxy)ethoxy)ethoxy)-7-methoxyquinazolin-4-yl)amino)-5-fluorophenol (7f).

The title compound was prepared according to the general procedure 5 using **5** (109 mg, 0.30 mmol) and 2-amino-4-chloro-5-fluorophenol (49 mg, 0.30 mmol) to obtain the product (117 mg, 80%) as a brown solid with impurities. It was used without further purification. MS-ESI *m/z* [M + H]⁺: calcd 487.3, found 486.3.

1⁷-methoxy-4,11-dioxa-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacycloundecaphane (8a).

The title compound was prepared according to the general procedure 7 using **6a** (93 mg, 0.23 mmol) to obtain the product (62 mg, 73%) as a orange solid. ¹H NMR (300 MHz, DMSO-d₆) δ 8.72 (s, 1H), 8.56 (s, 1H), 8.54 – 8.49 (m, 1H), 7.49 (s, 1H), 7.21 (s, 1H), 7.08 – 6.99 (m, 3H), 4.46 – 4.35 (m, 2H), 4.14 – 4.06 (m, 2H), 3.93 (s, 3H), 1.97 – 1.73 (m, 6H), 1.52 – 1.37 (m, 2H). ¹³C NMR (75 MHz, DMSO) δ 155.34, 154.68, 152.96, 148.77, 147.66, 145.74, 128.66, 123.20, 120.56, 120.39, 111.60, 108.87, 107.24, 100.96, 68.46, 67.30, 55.88, 29.59, 24.41, 24.24, 22.86. MS-ESI *m/z* [M + H]⁺: calcd 366.4, found 366.3. HRMS *m/z* [M + H]⁺: calcd 366.1812, found 366.1814. HPLC: t_R = 8.17, purity ≥ 95% (UV: 254/ 280 nm).

3³-chloro-1⁷-methoxy-4,11-dioxo-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacycloundecaphane (8b).

The title compound was prepared according to the general procedure 7 using **6b** (97 mg, 0.22 mmol) to obtain the product (54 mg, 61%) as a white solid. ¹H NMR (300 MHz, CD₂Cl₂) δ 9.03 (dd, J = 8.1, 1.7 Hz, 1H), 8.66 (s, 1H), 8.42 (s, 1H), 7.23 (s, 1H), 7.16 – 6.99 (m, 3H), 4.48 – 4.38 (m, 2H), 4.34 (t, J = 5.0 Hz, 2H), 3.99 (s, 3H), 2.07 – 1.83 (m, 6H), 1.72 – 1.58 (m, 2H). ¹³C NMR (75 MHz, CD₂Cl₂) δ 155.64, 155.56, 153.66, 149.20, 147.49, 145.94, 134.85, 126.44, 125.22, 123.70, 118.28, 109.76, 108.78, 99.43, 76.38, 67.56, 56.71, 31.98, 25.71, 25.58, 24.74. MS-ESI *m/z* [M + H]⁺: calcd 400.9, found 400.3. HRMS *m/z* [M + H]⁺: calcd 400.1423, found 400.1422. HPLC: t_R = 8.90, purity ≥ 95% (UV: 254/ 280 nm).

3⁴-chloro-1⁷-methoxy-4,11-dioxo-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacycloundecaphane (8c).

The title compound was prepared according to the general procedure 7 using **6c** (146 mg, 0.33 mmol) to obtain the product (63 mg, 47%) as a orange solid. ¹H NMR (300 MHz, CD₂Cl₂) δ 8.74 (d, J = 8.7 Hz, 1H), 8.63 (s, 1H), 8.41 (s, 1H), 7.20 (s, 1H), 7.16 (s, 1H), 7.03 (dd, J = 8.6, 2.3 Hz, 1H), 6.91 (d, J = 2.3 Hz, 1H), 4.40 – 4.29 (m, 2H), 4.14 (t, J = 4.5 Hz, 2H), 3.98 (s, 3H), 2.08 – 1.82 (m, 6H), 1.69 – 1.49 (m, 2H). ¹³C NMR (75 MHz, CD₂Cl₂) δ 155.59, 155.46, 154.05, 149.53, 148.77, 147.19, 128.84, 127.14, 121.52, 119.44, 112.18, 109.95, 108.62, 99.86, 69.70, 69.56, 56.71, 30.70, 25.65, 25.48, 23.77. MS-ESI *m/z* [M + H]⁺: calcd 400.9, found 400.4. HRMS *m/z* [M + H]⁺: calcd 400.1423, found HPLC: t_R = 8.80, purity ≥ 95% (UV: 254/ 280 nm).

3⁵-chloro-1⁷-methoxy-4,11-dioxo-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacycloundecaphane (8d).

The title compound was prepared according to the general procedure 7 using **6d** (162 mg, 0.37 mmol) to obtain the product (52 mg, 35%) as a brown solid. ¹H NMR (300 MHz, DMSO-d₆) δ 8.83 (s, 1H), 8.66 (s, 1H), 8.61 (d, J = 2.0 Hz, 1H), 7.48 (s, 1H), 7.23 (s, 1H), 7.12 – 7.07 (m, 2H), 4.40 (t, J = 8.4 Hz, 2H), 4.12 (t, 2H), 3.95 (s, 3H), 1.98 – 1.68 (m, 6H), 1.52 – 1.37 (m, 2H). ¹³C NMR (75 MHz, DMSO) δ 155.13, 155.03, 152.57, 147.93, 147.64, 129.72, 124.21, 122.50, 119.57, 113.03, 108.78, 106.74, 101.06, 69.10, 67.47, 55.99, 29.51, 24.39, 24.20, 22.85. MS-ESI *m/z* [M + H]⁺: calcd 400.9, found 400.4. HRMS *m/z* [M + H]⁺: calcd 400.1423, found 400.1423. HPLC: t_R = 8.91, purity ≥ 95% (UV: 254/ 280 nm).

3⁶-chloro-1⁷-methoxy-4,11-dioxo-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacycloundecaphane (8e).

The title compound was prepared according to the general procedure 7 using **6e** (77 mg, 0.18 mmol) to obtain the product (15 mg, 21%) as a beige solid. ¹H NMR (300 MHz, CD₂Cl₂) δ 8.53 (s, 1H), 7.37 (s, 1H), 7.25 – 7.10 (m, 4H), 6.84 (dd, J = 8.0, 1.5 Hz, 1H), 4.50 – 4.36 (m, 2H), 4.02 – 3.95 (m, 5H), 1.79 – 1.67 (m, 6H), 1.49 – 1.37 (m, 2H). ¹³C NMR (75 MHz, CD₂Cl₂) δ 158.55, 156.19, 154.72, 154.10, 148.19, 147.80, 133.84, 127.85, 127.74, 122.49, 109.99, 109.96, 108.52, 101.04, 68.89, 66.91, 56.68, 29.69, 25.58, 25.54,

23.66. MS-ESI m/z $[M + H]^+$: calcd 400.9, found 400.3. HRMS m/z $[M + H]^+$: calcd found 400.1421. HPLC: t_R = 7.67, purity $\geq$ 95% (UV: 254/ 280 nm).

3⁵-chloro-3⁴-fluoro-1⁷-methoxy-4,11-dioxa-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacycloundecaphane (8f).

The title compound was prepared according to the general procedure 7 using **6f** (132 mg, 0.29 mmol) to obtain the product (52 mg, 43%) as a purple solid. ¹H NMR (250 MHz, methylene chloride-d₂) δ 8.88 (d, J = 8.1 Hz, 1H), 8.64 (s, 1H), 8.23 (s, 1H), 7.18 (s, 1H), 7.07 (s, 1H), 6.73 (d, J = 10.0 Hz, 1H), 4.38 – 4.26 (m, 2H), 4.13 – 4.07 (m, 2H), 3.98 (s, 3H), 2.03 – 1.79 (m, 6H), 1.65 – 1.48 (m, 2H). ¹³C NMR (75 MHz, methylene chloride-d₂) δ 155.63, 155.20, 153.95, 153.47 (d, J = 242.7 Hz), 149.58, 147.55 (d, J = 8.8 Hz), 147.16, 126.92 (d, J = 3.2 Hz), 119.45, 112.11 (d, J = 18.0 Hz), 109.73, 108.60, 101.12 (d, J = 26.3 Hz), 99.65, 70.02, 69.57, 56.71, 30.58, 25.63, 25.47, 23.76. MS-ESI m/z $[M + H]^+$: calcd 418.9, found 418.4. HRMS m/z $[M + H]^+$: calcd 418.1328, found 418.1327. HPLC: t_R = 5.79, purity $\geq$ 95% (UV: 254/ 280 nm).

1⁷-methoxy-4,7,10,13-tetraoxa-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacyclotridecaphane (9a).

The title compound was prepared according to the general procedure 7 using **7a** (95 mg, 0.22 mmol) to obtain the product (84 mg, 100%) as a pale yellow solid. ¹H NMR (300 MHz, DMSO-d₆) δ 10.52 (s, 1H), 8.73 (s, 1H), 8.31 (s, 1H), 7.64 (d, J = 7.6 Hz, 1H), 7.41 – 7.32 (m, 2H), 7.18 (d, J = 8.0 Hz, 1H), 7.09 (t, J = 7.5 Hz, 1H), 4.66 – 4.58 (m, 2H), 4.16 – 4.08 (m, 2H), 3.99 (s, 3H), 3.79 – 3.72 (m, 2H), 3.66 – 3.59 (m, 2H), 3.56 – 3.49 (m, 2H), 3.48 – 3.43 (m, 2H). ¹³C NMR (75 MHz, DMSO) δ 162.72, 158.83, 156.57, 152.70, 149.72, 148.88, 128.42, 127.70, 125.43, 120.61, 113.15, 106.68, 106.47, 99.65, 70.62, 70.01, 69.62, 68.68, 68.29, 67.78, 56.38. MS-ESI m/z $[M + H]^+$: calcd 398.2, found 398.3. HRMS m/z $[M + H]^+$: calcd 398.1711, found 398.1705. HPLC: t_R = 7.03, purity $\geq$ 95% (UV: 254/ 280 nm).

3³-chloro-1⁷-methoxy-4,7,10,13-tetraoxa-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacyclotridecaphane (9b).

The title compound was prepared according to the general procedure 7 using **7b** (78 mg, 0.17 mmol) to obtain the product (49 mg, 68%) as a white solid. ¹H NMR (300 MHz, DMSO-d₆) δ 10.77 (s, 1H), 8.76 (s, 1H), 8.40 (s, 1H), 7.70 (d, J = 7.8 Hz, 1H), 7.49 (d, J = 7.9 Hz, 1H), 7.43 (s, 1H), 7.27 (t, J = 8.0 Hz, 1H), 4.64 – 4.55 (m, 2H), 4.02 – 3.94 (m, 5H), 3.84 – 3.76 (m, 2H), 3.64 – 3.56 (m, 2H), 3.53 – 3.45 (m, 2H), 3.45 – 3.37 (m, 2H). ¹³C NMR (75 MHz, DMSO) δ 158.63, 156.80, 149.38, 149.10, 149.03, 136.30, 132.11, 128.36, 127.17, 127.04, 124.90, 107.57, 106.88, 100.21, 72.82, 71.18, 70.13, 70.02, 69.23, 68.40, 56.37. MS-ESI m/z $[M + H]^+$: calcd 432.1, found 432.3. HRMS m/z $[M + H]^+$: calcd 432.1321, found 432.1319. HPLC: t_R = 7.49, purity $\geq$ 95% (UV: 254/ 280 nm).

3⁴-chloro-1⁷-methoxy-4,7,10,13-tetraoxa-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacyclotridecaphane (9c).

The title compound was prepared according to the general procedure 7 using **7c** (120 mg, 0.26 mmol) to obtain the product (86 mg, 72%) as a yellow solid. ¹H NMR (300 MHz, DMSO-d₆) δ 10.49 (s, 1H), 8.75 (s, 1H), 8.30 (s, 1H), 7.69 (d, J = 8.4 Hz, 1H), 7.43 (s, 1H), 7.29 (d, J = 2.1 Hz, 1H), 7.16 (dd, J = 8.5, 2.1 Hz, 1H), 4.67 – 4.58 (m, 2H), 4.20 – 4.12 (m, 2H), 3.98 (s, 3H), 3.80 – 3.72 (m, 2H), 3.65 – 3.58 (m, 2H), 3.56 – 3.49 (m, 2H), 3.47 – 3.40 (m, 2H). ¹³C NMR (75 MHz, DMSO) δ 158.77, 156.60, 153.35, 149.74, 148.87, 135.39, 132.02, 128.69, 124.65, 120.45, 113.53, 106.61, 106.57, 99.84, 70.55, 69.97, 69.57, 68.77, 68.47, 67.81, 56.37. MS-ESI *m/z* [M + H]⁺: calcd 432.1, found 432.3. HRMS *m/z* [M + H]⁺: calcd 432.1321, found 432.1319. HPLC: t_R = 7.48, purity ≥ 95% (UV: 254/ 280 nm).

3⁵-chloro-1⁷-methoxy-4,7,10,13-tetraoxa-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacyclotridecaphane (9d).

The title compound was prepared according to the general procedure 7 using **7d** (94 mg, 0.20 mmol) to obtain the product (68 mg, 79%) as a pale yellow solid. ¹H NMR (300 MHz, DMSO-d₆) δ 10.27 (s, 1H), 8.79 (s, 1H), 8.21 (s, 1H), 7.91 (d, J = 2.5 Hz, 1H), 7.38 (dd, J = 8.7, 2.5 Hz, 1H), 7.33 (s, 1H), 7.21 (d, J = 8.8 Hz, 1H), 4.59 (t, J = 4.6 Hz, 2H), 4.20 – 4.12 (m, 2H), 3.99 (s, 3H), 3.76 (t, J = 4.3 Hz, 2H), 3.71 – 3.64 (m, 2H), 3.59 – 3.52 (m, 2H), 3.53 – 3.46 (m, 2H). ¹³C NMR (75 MHz, DMSO) δ 158.20, 156.67, 155.74, 151.04, 149.74, 149.52, 135.88, 127.17, 124.47, 123.98, 119.13, 114.52, 106.86, 106.44, 70.37, 69.84, 69.68, 68.76, 68.53, 68.22, 56.38. MS-ESI *m/z* [M + H]⁺: calcd 432.1, found 432.3. HRMS *m/z* [M + H]⁺: calcd 432.1321, found 432.1319. HPLC: t_R = 7.44, purity ≥ 95% (UV: 254/ 280 nm).

3⁶-chloro-1⁷-methoxy-4,7,10,13-tetraoxa-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacyclotridecaphane (9e).

The title compound was prepared according to the general procedure 7 using **7e** (109 mg, 0.23 mmol) to obtain the product (61 mg, 60%) as a yellow solid. ¹H NMR (300 MHz, DMSO-d₆) δ 10.87 (s, 1H), 8.72 (s, 1H), 8.41 (s, 1H), 7.55 – 7.32 (m, 2H), 7.25 (d, J = 8.0 Hz, 1H), 7.17 (d, J = 8.1 Hz, 1H), 5.01 – 4.85 (m, 1H), 4.35 (d, J = 13.0 Hz, 1H), 4.20 (d, J = 10.2 Hz, 1H), 4.00 (s, 3H), 3.96 – 3.89 (m, 1H), 3.84 – 3.70 (m, 2H), 3.63 (t, J = 11.8 Hz, 2H), 3.50 – 3.38 (m, 2H), 3.28 – 3.17 (m, 2H). ¹³C NMR (75 MHz, DMSO) δ 160.93, 160.01, 156.70, 155.57, 149.86, 148.84, 133.07, 130.01, 123.75, 121.44, 112.05, 106.68, 106.16, 99.41, 71.20, 70.46, 69.56, 68.67, 68.59, 67.06, 56.41. MS-ESI *m/z* [M + H]⁺: calcd 432.1, found 432.3. HRMS *m/z* [M + H]⁺: calcd 432.1321, found 432.1319. HPLC: t_R = 7.12, purity ≥ 95% (UV: 254/ 280 nm).

3⁵-chloro-3⁴-fluoro-1⁷-methoxy-4,7,10,13-tetraoxa-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacyclotridecaphane (9f).

The title compound was prepared according to the general procedure 7 using **7f** (117 mg, 0.24 mmol) to obtain the product (6 mg, 6%) as a white solid. ¹H NMR (250 MHz, DMSO-d₆) δ 10.05 (s, 1H), 8.75 (s, 1H), 8.15 (s, 1H), 8.07 (d, J = 8.1 Hz, 1H), 7.39 (d, J = 11.0 Hz, 1H), 7.25 (s, 1H), 4.62 – 4.54 (m, 2H), 4.23 – 4.15 (m, 2H), 3.99 (s, 3H),

3.75 (t, $J = 4.4$ Hz, 2H), 3.71 – 3.63 (m, 2H), 3.55 – 3.52 (m, 4H). ^{13}C NMR (75 MHz, DMSO) δ 158.20, 157.36 (d, $J = 122.8$ Hz), 154.22 (d, $J = 16.3$ Hz), 152.16, 152.02, 150.06, 149.64, 127.35 (d, $J = 1.6$ Hz), 123.24 (d, $J = 3.6$ Hz), 109.89 (d, $J = 19.1$ Hz), 106.96, 106.20, 102.67 (d, $J = 25.9$ Hz), 101.57, 70.34, 69.88, 69.67, 69.16, 68.34, 68.20, 56.35. MS-ESI m/z $[\text{M} + \text{H}]^+$: calcd 450.9, found 450.3. HRMS m/z $[\text{M} + \text{H}]^+$: calcd 450.1227, found 450.1224. HPLC: $t_R = 7.52$, purity $\geq 95\%$ (UV: 254/ 280 nm).

4-chloro-6-(2-(2-chloroethoxy)ethoxy)-7-methoxyquinazoline (10).

The title compound was prepared according to the general procedure 4 using 2-(2-chloroethoxy)ethanol (185 mg, 1.48 mmol) and heated for 2 h to obtain the product (159 mg, 42%) as a yellow solid. ^1H NMR (300 MHz, DMSO- d_6) δ 8.87 (s, 1H), 7.45 (s, 1H), 7.43 (s, 1H), 4.45 – 4.29 (m, 2H), 4.02 (s, 3H), 3.94 – 3.87 (m, 2H), 3.82 – 3.73 (m, 4H). ^{13}C NMR (75 MHz, DMSO) δ 157.95, 156.79, 152.22, 150.55, 148.57, 118.54, 106.97, 103.31, 70.68, 68.55, 68.48, 56.54, 43.51. MS-ESI m/z $[\text{M} + \text{H}]^+$: calcd 317.1, found 317.1.

3-((6-(2-(2-chloroethoxy)ethoxy)-7-methoxyquinazolin-4-yl)amino)phenol (11a).

The title compound was prepared according to the general procedure 6 using **10** (157 mg, 0.50 mmol) and 3-aminophenol (55 mg, 0.50 mmol) to obtain the product (155 mg, 80%) as a pale yellow solid. ^1H NMR (300 MHz, DMSO- d_6) δ 11.29 (s, 1H), 9.74 (s, 1H), 8.80 (s, 1H), 8.37 (s, 1H), 7.40 (s, 1H), 7.25 (t, $J = 8.0$ Hz, 1H), 7.17 – 7.05 (m, 2H), 6.86 – 6.68 (m, 1H), 4.48 – 4.29 (m, 2H), 3.99 (s, 3H), 3.94 – 3.87 (m, 2H), 3.86 – 3.71 (m, 4H). ^{13}C NMR (75 MHz, DMSO) δ 158.11, 157.66, 156.25, 149.21, 148.59, 137.73, 135.41, 129.32, 115.43, 113.50, 111.91, 107.16, 104.89, 99.82, 70.71, 68.99, 68.36, 56.45, 43.56. MS-ESI m/z $[\text{M} + \text{H}]^+$: calcd 390.1, found 390.1. HRMS m/z $[\text{M} + \text{H}]^+$: calcd 390.1215, found 390.1211. HPLC: $t_R = 7.12$, purity $\geq 95\%$ (UV: 254/ 280 nm).

2-chloro-5-((6-(2-(2-chloroethoxy)ethoxy)-7-methoxyquinazolin-4-yl)amino)phenol (11b).

The title compound was prepared according to the general procedure 6 using **10** (200 mg, 0.63 mmol) and 5-amino-2-chlorophenol (91 mg, 0.63 mmol) to obtain the product (84 mg, 31%) as a pale green solid. ^1H NMR (300 MHz, DMSO- d_6) δ 11.23 (s, 1H), 10.53 (s, 1H), 8.80 (s, 1H), 8.34 (s, 1H), 7.64 – 7.35 (m, 3H), 7.17 (d, $J = 8.5$ Hz, 1H), 4.45 – 4.29 (m, 2H), 3.99 (s, 3H), 3.95 – 3.88 (m, 2H), 3.83 – 3.73 (m, 4H). ^{13}C NMR (75 MHz, DMSO) δ 157.94, 156.17, 153.05, 149.14, 148.90, 136.60, 129.47, 117.01, 116.34, 112.76, 107.35, 104.74, 100.38, 70.69, 68.96, 68.35, 56.42, 43.54. MS-ESI m/z $[\text{M} + \text{H}]^+$: calcd 424.1, found 424.0. HRMS m/z $[\text{M} + \text{H}]^+$: calcd 424.0825, found 424.0826. HPLC: $t_R = 7.48$, purity $\geq 95\%$ (UV: 254/ 280 nm).

3-chloro-5-((6-(2-(2-chloroethoxy)ethoxy)-7-methoxyquinazolin-4-yl)amino)phenol (11c).

The title compound was prepared according to the general procedure 6 using **10** (200 mg, 0.63 mmol) and 3-amino-5-chlorophenol (91 mg, 0.63 mmol) to obtain the product (89 mg, 33%) as a pale yellow solid. ^1H NMR (300 MHz, DMSO- d_6) δ 11.31 (s, 1H), 10.28 (s, 1H), 8.88 (s, 1H), 8.37 (s, 1H), 7.38 (s, 1H), 7.31 (t, $J = 1.9$ Hz, 1H), 7.19 (t, $J = 2.0$ Hz, 1H), 6.77 (t, $J = 2.0$ Hz, 1H), 4.38 (t, $J = 4.5$ Hz, 2H), 4.00 (s, 3H), 3.96 – 3.88 (m, 2H), 3.86 – 3.73 (m, 4H). ^{13}C NMR (75 MHz, DMSO) δ 158.57, 158.15, 156.46, 149.33, 148.71,

138.99, 135.72, 133.06, 115.01, 113.07, 110.43, 107.36, 104.84, 99.90, 70.72, 69.05, 68.36, 56.51, 43.55. MS-ESI m/z $[M + H]^+$: calcd 424.1, found 424.0. HRMS m/z $[M + H]^+$: calcd 424.0825, found 424.0827. HPLC: t_R = 7.52, purity $\geq$ 95% (UV: 254/ 280 nm).

4-chloro-3-((6-(2-(2-chloroethoxy)ethoxy)-7-methoxyquinazolin-4-yl)amino)phenol (11d).

The title compound was prepared according to the general procedure 5 using **10** (157 mg, 0.50 mmol) and 3-amino-4-chlorophenol (71 mg, 0.50 mmol) to obtain the product (70 mg, 33%) as a pale green solid. 1H NMR (300 MHz, DMSO- d_6) δ 11.44 (s, 1H), 10.12 (s, 1H), 8.76 (s, 1H), 8.31 (s, 1H), 7.52 – 7.32 (m, 2H), 6.96 (d, J = 2.6 Hz, 1H), 6.86 (dd, J = 8.7, 2.7 Hz, 1H), 4.40 – 4.31 (m, 2H), 4.00 (s, 3H), 3.95 – 3.86 (m, 2H), 3.86 – 3.72 (m, 4H). ^{13}C NMR (75 MHz, DMSO) δ 159.10, 156.98, 156.43, 149.30, 148.97, 136.03, 134.62, 130.24, 120.24, 116.30, 116.19, 106.71, 104.56, 100.18, 70.68, 68.84, 68.32, 56.49, 43.54. MS-ESI m/z $[M + H]^+$: calcd 424.1, found 424.1. HRMS m/z $[M + H]^+$: calcd 424.0825, found 424.0827. HPLC: t_R = 7.34, purity $\geq$ 95% (UV: 254/ 280 nm).

1^7 -methoxy-4,7,10-trioxa-2-aza-1(4,6)-quinazolina-3(1,3)-benzenacyclodecaphane (12a).

The title compound was prepared according to the general procedure 7 using **11a** (135 mg, 0.35 mmol) to obtain the product (22 mg, 18%) as a yellow solid. 1H NMR (300 MHz, DMSO- d_6) δ 9.53 (s, 1H), 8.50 (s, 1H), 7.25 (t, J = 7.8 Hz, 1H), 7.16 (s, 1H), 7.03 (s, 1H), 6.86 – 6.70 (m, 3H), 4.22 – 4.12 (m, 2H), 4.04 – 3.94 (m, 2H), 3.90 (s, 3H), 3.60 – 3.46 (m, 4H). ^{13}C NMR (75 MHz, DMSO) δ 159.92, 154.74, 153.11, 148.79, 146.67, 141.77, 129.75, 115.74, 113.62, 109.81, 109.15, 107.02, 106.95, 71.75, 70.89, 66.83, 66.60, 55.70, 29.24. MS-ESI m/z $[M + H]^+$: calcd 354.1, found 354.3. HRMS m/z $[M + H]^+$: calcd 354.1448, found 354.1449. HPLC: t_R = 8.86, purity $\geq$ 95% (UV: 254/ 280 nm).

3^4 -chloro- 1^7 -methoxy-4,7,10-trioxa-2-aza-1(4,6)-quinazolina-3(1,3)-benzenacyclodecaphane (12b).

The title compound was prepared according to the general procedure 7 using **11b** (74 mg, 0.17 mmol) to obtain the product (20 mg, 30%) as a yellow solid. 1H NMR (250 MHz, DMSO- d_6) δ 11.32 (s, 1H), 8.82 (s, 1H), 7.75 – 7.29 (m, 3H), 7.00 (s, 1H), 6.85 (s, 1H), 4.68 – 4.19 (m, 2H), 3.92 (s, 3H), 3.64 – 3.42 (m, 6H). ^{13}C NMR (75 MHz, DMSO) δ 158.73, 155.84, 155.74, 147.68, 147.35, 137.06, 136.55, 130.09, 121.97, 119.01, 115.63, 106.22, 105.47, 99.47, 72.17, 71.06, 68.77, 66.74, 56.02. MS-ESI m/z $[M + H]^+$: calcd 388.1, found 388.1. HRMS m/z $[M + H]^+$: calcd 388.1059, found 388.1059. HPLC: t_R = 7.24, purity $\geq$ 95% (UV: 254/ 280 nm).

3^5 -chloro- 1^7 -methoxy-4,7,10-trioxa-2-aza-1(4,6)-quinazolina-3(1,3)-benzenacyclodecaphane (12c).

The title compound was prepared according to the general procedure 7 using **11c** (79 mg, 0.19 mmol) to obtain the product (6 mg, 8%) as a white solid. 1H NMR (300 MHz, DMSO- d_6) δ 9.66 (s, 1H), 8.59 (s, 1H), 7.22 (s, 1H), 7.12 (s, 1H), 6.83 (t, J = 1.8 Hz, 1H), 6.76 – 6.63 (m, 2H), 4.20 – 4.09 (m, 4H), 3.93 (s, 3H), 3.62 – 3.51 (m, 4H). ^{13}C NMR (75 MHz, DMSO) δ 160.54, 155.12, 155.09, 152.68, 148.32, 147.39, 147.31, 133.41, 113.61, 112.28, 110.74, 109.82, 106.71, 106.24, 71.69, 66.74, 66.37, 55.82, 54.90. MS-ESI m/z $[M$

+ H]⁺: calcd 388.1, found 388.1. HRMS *m/z* [M + H]⁺: calcd 388.1059, found 388.1059.
HPLC: *t_R* = 7.35, purity ≥ 95% (UV: 254/ 280 nm).

3⁶-chloro-1⁷-methoxy-4,7,10-trioxa-2-aza-1(4,6)-quinazolina-3(1,3)-benzenacyclodecaphane (12d).

The title compound was prepared according to the general procedure 7 using **11d** (60 mg, 0.14 mmol) to obtain the product (15 mg, 27%) as a pale yellow solid. ¹H NMR (300 MHz, DMSO-*d*₆) δ 9.51 (s, 1H), 8.44 (s, 1H), 7.45 (d, *J* = 8.8 Hz, 1H), 7.18 – 7.12 (m, 2H), 6.95 (dd, *J* = 8.8, 2.8 Hz, 1H), 6.74 (s, 1H), 4.30 – 4.21 (m, 2H), 3.91 – 3.84 (m, 5H), 3.63 – 3.55 (m, 2H), 3.45 (t, *J* = 4.8 Hz, 2H). ¹³C NMR (75 MHz, DMSO) δ 159.53, 157.11, 154.48, 152.57, 148.50, 146.82, 138.79, 130.01, 121.33, 117.29, 116.10, 108.75, 107.31, 106.44, 71.94, 70.05, 68.19, 67.20, 55.68. MS-ESI *m/z* [M + H]⁺: calcd 388.1, found HRMS *m/z* [M + H]⁺: calcd 388.1059, found 388.1059. HPLC: *t_R* = 7.09, purity ≥ 95% (UV: 254/ 280 nm).

4-oxo-3,4-dihydroquinazolin-6-yl acetate (14).

A mixture of 6-hydroxyquinazolin-4(3H)-one (5.58 g, 34.4 mmol) and pyridine (33.3 mL, 0.41 mol) in 32.5 mL acetic anhydride was stirred at 50 °C for 1 h. Ice was added to the reaction mixture after cooling it down to rt, leading to the precipitation of **14**. It was filtered and washed with ice water to obtain the product (7.02 g, 85%) as a beige solid. ¹H NMR (250 MHz, DMSO-*d*₆) δ 12.31 (s, 1H), 8.09 (s, 1H), 7.83 (d, *J* = 2.7 Hz, 1H), 7.72 (d, *J* = 8.8 Hz, 1H), 7.58 (ddd, *J* = 8.8, 2.7, 0.6 Hz, 1H), 2.31 (s, 3H). MS-ESI *m/z* [M + H]⁺: calcd 203.2, found 203.2.

4-chloroquinazolin-6-yl acetate (15).

A mixture of **14** (5.97 g, 29.2 mmol) and thionyl chloride (45.0 mL, 0.62 mol) in 0.4 mL anhydrous DMF was stirred at 70 °C for 3 h. The solvent was evaporated under reduced pressure and the crude product was used without further purification. ¹H NMR (250 MHz, chloroform-*d*) δ 9.05 (s, 1H), 8.11 (d, *J* = 9.1 Hz, 1H), 8.02 (d, *J* = 2.5 Hz, 1H), 7.73 (dd, *J* = 9.1, 2.5 Hz, 1H), 2.40 (s, 3H). MS-ESI *m/z* [M + H]⁺: calcd 245.0, found 245.0.

4-chloroquinazolin-6-ol (16).

A mixture of **15** (6.51 g, 29.2 mmol) in 132 mL ammonia in methanol (2 M) was stirred at rt for 1 h. The solvent was evaporated under reduced pressure and the crude product was purified by flash chromatography using n-hexane/ ethyl acetate as an eluent to obtain the product (1.00 g, 19%) as a beige solid. ¹H NMR (250 MHz, DMSO-*d*₆) δ 10.84 (s, 0H), 8.89 (s, 1H), 7.98 (d, *J* = 9.1 Hz, 1H), 7.64 (dd, *J* = 9.1, 2.7 Hz, 1H), 7.42 (d, *J* = 2.7 Hz, 1H). MS-ESI *m/z* [M + H]⁺: calcd 181.6, found 181.1.

4-((5-chloro-4-fluoro-2-hydroxyphenyl)amino)quinazolin-6-ol (17).

16 (154 mg, 0.83 mmol) and 2-amino-4-chloro-5-fluorophenol (148 mg, 0.91 mmol) were dissolved in 11 mL anhydrous ethanol. The mixture was stirred at 70 °C for 18 h. A solid participated, which was filtered and washed with ethanol to obtain the product (190 mg, 75%) as a yellow solid. ¹H NMR (500 MHz, DMSO-*d*₆) δ 11.11 (s, 1H), 10.95 (s, 1H), 10.84 (s, 1H), 8.76 (s, 1H), 7.95 (d, *J* = 2.5 Hz, 1H), 7.89 (d, *J* = 9.1 Hz, 1H), 7.71 (dd, *J* =

9.1, 2.5 Hz, 1H), 7.61 (d, $J = 8.2$ Hz, 1H), 7.09 (d, $J = 10.9$ Hz, 1H). ^{13}C NMR (126 MHz, DMSO) δ 158.89 (d, $J = 358.2$ Hz), 157.83, 155.51, 153.28 (d, $J = 10.6$ Hz), 148.22, 131.65, 129.43, 126.94, 121.29, 121.22 (d, $J = 3.0$ Hz), 114.69, 108.22 (d, $J = 18.9$ Hz), 107.06, 104.76 (d, $J = 23.8$ Hz). MS-ESI m/z $[\text{M} + \text{H}]^+$: calcd 306.7, found 306.4. HRMS m/z $[\text{M} + \text{H}]^+$: calcd 306.0440, found 306.0441. HPLC: $t_R = 6.91$, purity $\geq 95\%$ (UV: 254/ 280 nm).

3⁵-chloro-3⁴-fluoro-4,7,10-trioxa-2-aza-1(4,6)-quinazolina-3(1,2)-benzenacyclodecaphane (18).

TPP (129 mg, 0.49 mmol) and DIAD (99 mg, 0.49 mmol) were dissolved in 41 mL anhydrous toluene. After 15 minutes at rt **17** (50 mg, 0.16 mmol) in 4 mL anhydrous THF was added. Another 15 minutes later 2,2'-oxydiethanol (17 mg, 0.16 mmol) was added and the mixture was stirred at 40 °C for 20 h. The solvent was evaporated under reduced pressure and the crude product was purified by flash chromatography using DCM/ methanol and *n*-hexane/ ethyl acetate as an eluent to obtain the product (3 mg, 5%) as a white solid. ^1H NMR (500 MHz, DMSO- d_6) δ 8.99 (s, 1H), 8.72 (s, 1H), 8.59 (d, $J = 8.1$ Hz, 1H), 8.45 (d, $J = 2.8$ Hz, 1H), 7.76 (d, $J = 9.0$ Hz, 1H), 7.55 – 7.47 (m, 2H), 4.61 – 4.56 (m, 2H), 4.33 – 4.29 (m, 2H), 3.97 – 3.92 (m, 2H), 3.92 – 3.87 (m, 2H). ^{13}C NMR (126 MHz, DMSO) δ 157.07, 156.57, 152.66 (d, $J = 243.4$ Hz), 152.46, 147.99, 144.20 (d, $J = 10.2$ Hz), 128.84 (d, $J = 6.1$ Hz), 125.62, 121.90 (d, $J = 17.7$ Hz), 119.24, 112.39 (d, $J = 18.6$ Hz), 116.02, 106.66 (d, $J = 24.8$ Hz), 104.47, 72.16, 71.56, 70.17, 69.06. MS-ESI m/z $[\text{M} + \text{H}]^+$: calcd 376.8, found 376.0. HRMS m/z $[\text{M} + \text{H}]^+$: calcd 376.0859, found 376.0867. HPLC: $t_R = 8.64$, purity $\geq 95\%$ (UV: 254/ 280 nm).

General procedure 1.

4-chloro-7-methoxyquinazolin-6-ol (1.0 equiv) and the corresponding aminophenol (1.1 equiv) were dissolved in anhydrous ethanol (0.07 M). The mixture was stirred at 70 °C for 17 h. A solid participated, which was filtered and washed with ethanol to obtain the title compound.

General procedure 2.

4-chloro-7-methoxyquinazolin-6-ol (1.0 equiv) and the corresponding aminophenol (1.1 equiv) were dissolved in anhydrous ethanol (0.07 M). The mixture was stirred at 70 °C for 17 h. The solvent was evaporated under reduced pressure and the crude product was purified by flash chromatography using DCM/ methanol as an eluent.

General procedure 3.

TPP (3.0 equiv) and DIAD (3.0 equiv) were dissolved in anhydrous toluene (0.01 M). After 10 minutes at rt the corresponding product of general procedure 1 or 2 (1.0 equiv) in anhydrous THF (0.05 M) was added. After additional 20 minutes 2,2'-oxydiethanol (1.0 equiv) was added and the mixture was stirred at 40 °C for 20 h. The solvent was evaporated under reduced pressure and the crude product was purified by flash chromatography using DCM/ methanol and *n*-hexane/ ethyl acetate as an eluent.

General procedure 4.

TPP (2.5 equiv) and DIAD (2.5 equiv) were dissolved in anhydrous toluene (0.05 M). After 15 minutes at rt the desired linker (1.25 equiv) was added. After additional 20 minutes 4-chloro-7-methoxyquinazolin-6-ol (1.0 equiv) was added and the mixture was stirred at 80 °C for 2 - 5 h. The solvent was evaporated under reduced pressure and the crude product was purified by flash chromatography using DCM/ methanol and *n*-hexane/ ethyl acetate as an eluent.

General procedure 5.

The corresponding product of general procedure 4 (1.0 equiv) and the desired aminophenol (1.0 equiv) were dissolved in anhydrous ethanol (0.04 M). The mixture was stirred at 70 °C for 18 h. The solvent was evaporated under reduced pressure and the crude product was purified by flash chromatography using DCM/ methanol as an eluent.

General procedure 6.

The corresponding product of general procedure 4 (1.0 equiv) and the desired aminophenol (1.0 equiv) were dissolved in anhydrous ethanol (0.04 M). The mixture was stirred at 70 °C for 18 h. A solid participated, which was filtered and washed with ethanol to obtain the title compound.

General procedure 7.

To a stirred solution of the corresponding product of general procedure 5 or 6 (1.0 equiv) in anhydrous DMF (0.002 M) was added NaH (3.0 equiv, 60% dispersion in oil) at 0 °C. The mixture was stirred for 10 minutes at 0 °C and another 24 h at 60 °C. The reaction was quenched with methanol and H₂O and was neutralized with a 4 M HCl solution. The crude product was purified by flash chromatography using DCM/ methanol as an eluent.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Abbreviations

DCM	dichloromethane
D	aspartic acid

DIAD	diisopropyl azodicarboxylate
DMF	dimethylformamide
DSF	differential scanning fluorimetry
F	phenylalanine
G	glycine
HTFR	Homogeneous time resolved fluorescence
M	methionine
NaH	sodium hydride
PBS	phosphate-buffered saline
rt	room temperature
SGF	simulated gastric fluid
SIF	simulated intestinal fluid
THF	tetrahydrofuran
TK	tyrosine kinase
TKI	tyrosine kinase inhibitor
TPP	triphenylphosphine
WT	wild type

References

- (1). Sung H; Ferlay J; Siegel RL; Laversanne M; Soerjomataram I; Jemal A; Bray F Global Cancer Statistics 2020: GLOBOCAN Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries. *CA. Cancer J. Clin* 2021, 71 (3), 209–249. [PubMed: 33538338]
- (2). Molina JR; Yang P; Cassivi SD; Schild SE; Adjei AA Non-Small Cell Lung Cancer: Epidemiology, Risk Factors, Treatment, and Survivorship. *Mayo Clin. Proc* 2008, 83 (5), 584–594. [PubMed: 18452692]
- (3). Bunn PA; Franklin W Epidermal Growth Factor Receptor Expression, Signal Pathway, and Inhibitors in Non-Small Cell Lung Cancer. *Semin. Oncol* 2002, 29 (5), 38–44.
- (4). Tamirat MZ; Koivu M; Elenius K; Johnson MS Structural Characterization of EGFR Exon 19 Deletion Mutation Using Molecular Dynamics Simulation. *PLoS One* 2019, 14 (9), 1–23.
- (5). Arteaga CL Overview of Epidermal Growth Factor Receptor Biology and Its Role as a Therapeutic Target in Human Neoplasia. *Semin. Oncol* 2002, 29 (5), 3–9.
- (6). Kosaka T; Yatabe Y; Endoh H; Kuwano H; Takahashi T; Mitsudomi T Mutations of the Epidermal Growth Factor Receptor Gene in Lung Cancer: Biological and Clinical Implications. *Cancer Res.* 2004, 64 (24), 8919–8923. [PubMed: 15604253]
- (7). Yun C-H; Boggen TJ; Li Y; Woo MS; Greulich H; Meyerson M; Eck MJ Structures of Lung Cancer-Derived EGFR Mutants and Inhibitor Complexes: Mechanism of Activation and Insights into Differential Inhibitor Sensitivity. *Cancer Cell.* 2007, 11 (3), 217–227. [PubMed: 17349580]

- (8). Williams R; Sanghera J; Wu F; Carbonaro-Hall D; Campbell DL; Warburton D; Pelech S; Hall F Identification of a Human Epidermal Growth Factor Receptor-Associated Protein Kinase as a New Member of the Mitogen-Activated Protein Kinase/Extracellular Signal-Regulated Protein Kinase Family. *J. Biol. Chem* 1993, 268 (24), 18213–18217. [PubMed: 7688742]
- (9). Phillips RJ; Mestas J; Gharaee-Kermani M; Burdick MD; Sica A; Belperio JA; Keane MP; Strieter RM Epidermal Growth Factor and Hypoxia-Induced Expression of CXC Chemokine Receptor 4 on Non-Small Cell Lung Cancer Cells Is Regulated by the Phosphatidylinositol 3-Kinase/PTEN/AKT/Mammalian Target of Rapamycin Signaling Pathway and Activation of Hypoxia Ind. *J. Biol. Chem* 2005, 280 (23), 22473–22481. [PubMed: 15802268]
- (10). Nakamura N; Chin H; Miyasaka N; Miura O An Epidermal Growth Factor Receptor/Jak2 Tyrosine Kinase Domain Chimera Induces Tyrosine Phosphorylation of Stat5 and Transduces a Growth Signal in Hematopoietic Cells. *J. Biol. Chem* 1996, 271 (32), 19483–19488. [PubMed: 8702638]
- (11). Ahmed SM; Salgia R Epidermal Growth Factor Receptor Mutations and Susceptibility to Targeted Therapy in Lung Cancer. *Respirology* 2006, 11 (6), 687–692. [PubMed: 17052295]
- (12). Hsu WH; Yang JCH; Mok TS; Loong HH Overview of Current Systemic Management of EGFR-Mutant NSCLC. *Ann. Oncol* 2018, 29 (Supplement 1), i3–i9. [PubMed: 29462253]
- (13). Paez JG; Jänne PA; Lee JC; Tracy S; Greulich H; Gabriel S; Herman P; Kaye FJ; Lindeman N; Boggon TJ; Naoki K; Sasaki H; Fujii Y; Eck MJ; Sellers WR; Johnson BE; Meyerson M EGFR Mutations in Lung Cancer: Correlation with Clinical Response to Gefitinib Therapy. *Science* (80-.) 2004, 304 (5676), 1497–1500.
- (14). Carey KD; Garton AJ; Romero MS; Kahler J; Thomson S; Ross S; Park F; Haley JD; Gibson N; Sliwkowski MX Kinetic Analysis of Epidermal Growth Factor Receptor Somatic Mutant Proteins Shows Increased Sensitivity to the Epidermal Growth Factor Receptor Tyrosine Kinase Inhibitor, Erlotinib. *Cancer Res.* 2006, 66 (16), 8163–8172. [PubMed: 16912195]
- (15). Harrison PT; Vyse S; Huang PH Rare Epidermal Growth Factor Receptor (EGFR) Mutations in Non-Small Cell Lung Cancer. *Semin. Cancer Biol* 2020, 61, 167–179. [PubMed: 31562956]
- (16). Kobayashi Y; Mitsudomi T Not All Epidermal Growth Factor Receptor Mutations in Lung Cancer Are Created Equal: Perspectives for Individualized Treatment Strategy. *Cancer Sci.* 2016, 107 (9), 1179–1186. [PubMed: 27323238]
- (17). Gridelli C; De Marinis F; Di Maio M; Cortinovis D; Cappuzzo F; Mok T Gefitinib as First-Line Treatment for Patients with Advanced Non-Small-Cell Lung Cancer with Activating Epidermal Growth Factor Receptor Mutation: Review of the Evidence. *Lung Cancer* 2011, 71 (3), 249–257. [PubMed: 21216486]
- (18). Nurwidya F; Takahashi F; Takahashi K Gefitinib in the Treatment of Nonsmall Cell Lung Cancer with Activating Epidermal Growth Factor Receptor Mutation. *J. Nat. Sci. Biol. Med* 2016, 7 (2), 119–123. [PubMed: 27433059]
- (19). Harvey RD; Adams VR; Beardslee T; Medina P Afatinib for the Treatment of EGFR Mutation-Positive NSCLC: A Review of Clinical Findings. *J. Oncol. Pharm. Pract* 2020, 26 (6), 1461–1474. [PubMed: 32567494]
- (20). Davis MI; Hunt JP; Herrgard S; Ciceri P; Wodicka LM; Pallares G; Hocker M; Treiber DK; Zarrinkar PP Comprehensive Analysis of Kinase Inhibitor Selectivity. *Nat. Biotechnol* 2011, 29 (11), 1046–1051. [PubMed: 22037378]
- (21). Blakely CM; Bivona TG Resiliency of Lung Cancers to EGFR Inhibitor Treatment Unveiled, Offering Opportunities to Divide and Conquer EGFR Inhibitor Resistance. *Cancer Discov.* 2012, 2 (10), 872–875. 10.1158/2159-8290.CD-12-0387. [PubMed: 23071030]
- (22). Gazdar AF Activating and Resistance Mutations of EGFR in Non-Small-Cell Lung Cancer: Role in Clinical Response to EGFR Tyrosine Kinase Inhibitors. *Oncogene* 2009, 28 (Suppl 1), S24–S31. [PubMed: 19680293]
- (23). Ward RA; Anderton MJ; Ashton S; Bethel PA; Box M; Butterworth S; Colclough N; Chorley CG; Chuaqui C; Cross DAE; Dakin LA; Debreczeni JÉ; Eberlein C; Finlay MRV; Hill GB; Grist M; Klinowska TCM; Lane C; Martin S; Orme JP; Smith P; Wang F; Waring MJ Structure- and Reactivity-Based Development of Covalent Inhibitors of the Activating and Gatekeeper Mutant Forms of the Epidermal Growth Factor Receptor (EGFR). *J. Med. Chem* 2013, 56 (17), 7025–7048. [PubMed: 23930994]

- (24). Wang S; Cang S; Liu D Third-Generation Inhibitors Targeting EGFR T790M Mutation in Advanced Non-Small Cell Lung Cancer. *J. Hematol. Oncol* 2016, 9 (1), 1–7. [PubMed: 26733151]
- (25). Remon J; Steuer CE; Ramalingam SS; Felip E Osimertinib and Other Third-Generation EGFR TKI in EGFR-Mutant NSCLC Patients. *Ann. Oncol* 2018, 29 (Supplement 1), i20–i27. [PubMed: 29462255]
- (26). Thress KS; Paweletz CP; Felip E; Cho BC; Stetson D; Dougherty B; Lai Z; Markovets A; Vivancos A; Kuang Y; Ercan D; Matthews SE; Cantarini M; Barrett JC; Jänne PA; Oxnard GR Acquired EGFR C797S Mutation Mediates Resistance to AZD9291 in Non-Small Cell Lung Cancer Harboring EGFR T790M. *Nat. Med* 2015, 21 (6), 560–562. [PubMed: 25939061]
- (27). Yang Z; Yang N; Ou Q; Xiang Y; Jiang T; Wu X; Bao H; Tong X; Wang X; Shao YW; Liu Y; Wang Y; Zhou C Investigating Novel Resistance Mechanisms to Third-Generation EGFR Tyrosine Kinase Inhibitor Osimertinib in Non-Small Cell Lung Cancer Patients. *Clin. Cancer Res* 2018, 24 (13), 3097–3107. [PubMed: 29506987]
- (28). Engelhardt H; Böse D; Petronczki M; Scharn D; Bader G; Baum A; Bergner A; Chong E; Döbel S; Egger G; Engelhardt C; Ettmayer P; Fuchs JE; Gerstberger T; Gonnella N; Grimm A; Grondal E; Haddad N; Hopfgartner B; Kousek R; Krawiec M; Kriz M; Lamarre L; Leung J; Mayer M; Patel ND; Simov BP; Reeves JT; Schnitzer R; Schrenk A; Sharps B; Solca F; Stadtmüller H; Tan Z; Wunberg T; Zoephel A; McConnell DB Start Selective and Rigidify: The Discovery Path toward a Next Generation of EGFR Tyrosine Kinase Inhibitors. *J. Med. Chem* 2019, 62 (22), 10272–10293. 10.1021/acs.jmedchem.9b01169. [PubMed: 31689114]
- (29). Collins JC; James K Emac - A Comparative Index for the Assessment of Macrocyclization Efficiency. *Medchemcomm* 2012, 3 (12), 1489–1495.
- (30). Amrhein JA; Knapp S; Hanke T Synthetic Opportunities and Challenges for Macrocyclic Kinase Inhibitors. *J. Med. Chem* 2021, 64 (12), 7991–8009. [PubMed: 34076436]
- (31). Lyssikatos JP; Marmsater FP; Zhao Q; Greschuk JM WO 2007/059257 A2, 2007.
- (32). Fedorov O; Niesen FH; Knapp S Kinase Inhibitor Selectivity Profiling Using Differential Scanning Fluorimetry. *Methods Mol. Biol* 2012, 795, 109–118. [PubMed: 21960218]
- (33). Blakely CM; Bivona TG Resiliency of Lung Cancers to EGFR Inhibitor Treatment Unveiled Offering Opportunities to Divide and Conquer EGFR Inhibitor Resistance. *Cancer Discov.* 2012, 2 (10), 872–875. [PubMed: 23071030]
- (34). Joshi SC Sol-Gel Behavior of Hydroxypropyl Methylcellulose (HPMC) in Ionic Media Including Drug Release. *Materials (Basel)*. 2011, 4 (10), 1861–1905. [PubMed: 28824113]
- (35). Dai L; Luo CY; Hu GX; Chen G; Wu CX; Yin J; Jiang ZY; Hu GF; Zhao J; Fu WF Comparative Analysis of First-Line Treatment Regimens for Advanced EGFR-Mutant Non-Small Cell Lung Cancer Patients with Stable Brain Metastases. *Ann. Cardiothorac. Surg* 2020, 9 (4), 2062–2071.
- (36). Herbst RS; LoRusso PM; Purdom M; Ward D Dermatologic Side Effects Associated with Gefitinib Therapy: Clinical Experience and Management. *Clin. Lung Cancer* 2003, 4 (6), 366–369. [PubMed: 14599302]
- (37). Fabbrocini G; Panariello L; Caro G; Cacciapuoti S Acneiform Rash Induced by EGFR Inhibitors: Review of the Literature and New Insights. *Ski. Appendage Disord* 2015, 1 (1), 31–37.
- (38). Jia Y; Yun CH; Park E; Ercan D; Manuia M; Juarez J; Xu C; Rhee K; Chen T; Zhang H; Palakurthi S; Jang J; Lelais G; DiDonato M; Bursulaya B; Michellys PY; Epple R; Marsilje TH; McNeill M; Lu W; Harris J; Bender S; Wong KK; Jänne PA; Eck MJ Overcoming EGFR(T790M) and EGFR(C797S) Resistance with Mutant-Selective Allosteric Inhibitors. *Nature* 2016, 534 (7605), 129–132. [PubMed: 27251290]
- (39). Rangachari D; To C; Shpilsky JE; Vanderlaan PA; Kobayashi SS; Mushajiang M; Lau CJ; Paweletz CP; Oxnard GR; Jänne PA; Costa DB EGFR-Mutated Lung Cancers Resistant to Osimertinib through EGFR-C797S Respond to 1st Generation Reversible EGFR Inhibitors but Eventually Acquire EGFR-T790M/C797S in Preclinical Models and Clinical Samples. *J Thorac Oncol.* 2019, 14 (11), 1995–2002. [PubMed: 31377341]
- (40). Kosaka T; Tanizaki J; Paranal RM; Endoh H; Lydon C; Capelletti M; Repellin CE; Choi J; Ogino A; Calles A; Ercan D; Redig AJ; Bahcall M; Oxnard GR; Eck MJ; Jänne PA Response

Heterogeneity of EGFR and HER2 Exon 20 Insertions to Covalent EGFR and HER2 Inhibitors. *Cancer Res.* 2017, 77 (10), 2712–2721. [PubMed: 28363995]

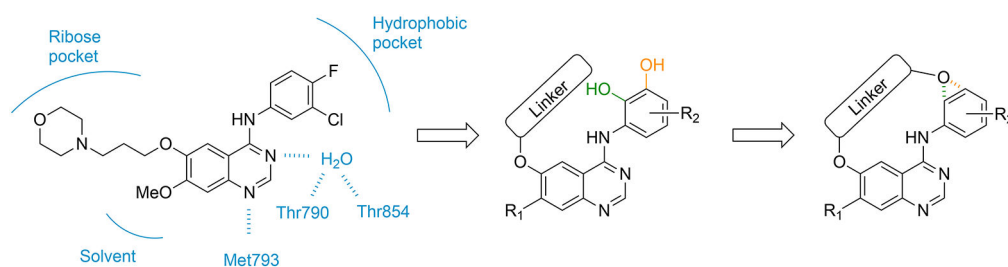
- (41). Zhou W; Ercan D; Chen L; Yun C; Li D; Capelletti M; Cortot AB; Chirieac L; Iacob RE; Padera R; Engen JR; Wong K; Eck MJ; Gray NS; Jänne PA Novel Mutant-Selective EGFR Kinase Inhibitors against EGFR T790M. *Nature* 2009, 462 (7276), 1070–1074. [PubMed: 20033049]

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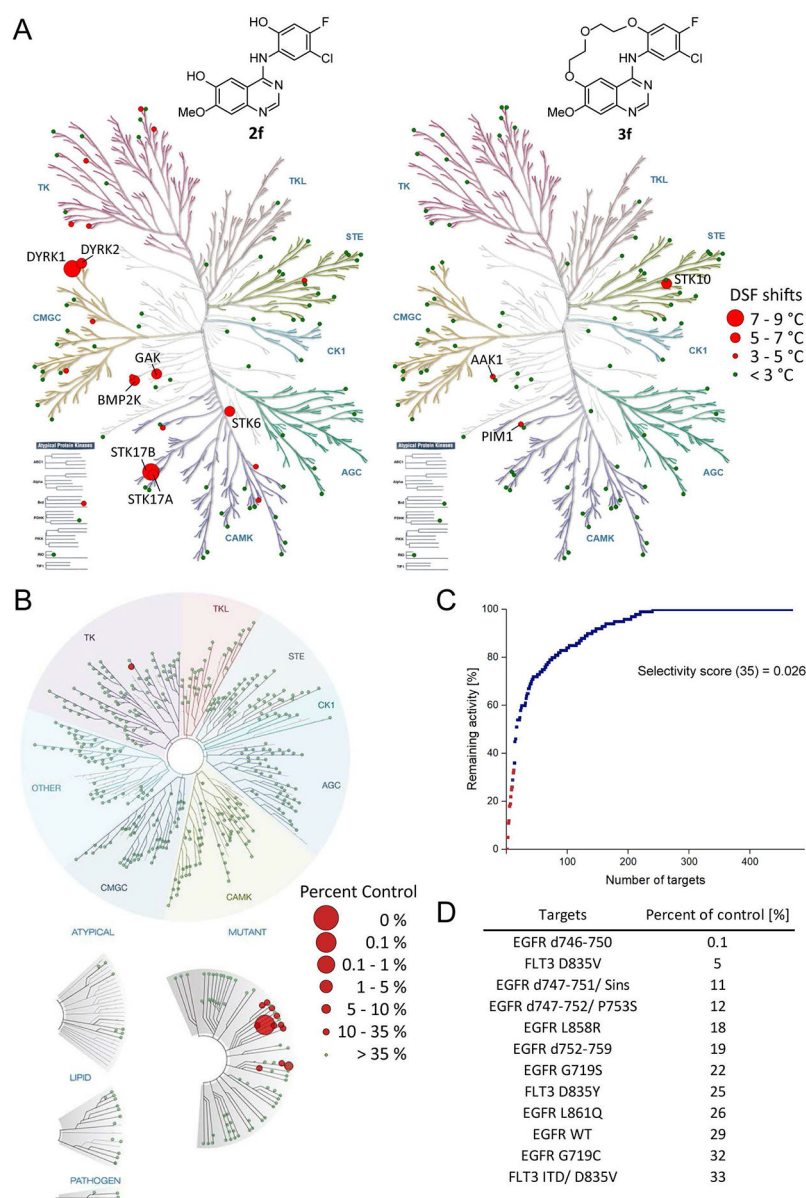
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**Figure 1:**

Synthetic strategy of the macrocyclic inhibitors starting from gefitinib (left). The different subunits and the polar interactions with the ATP binding pocket of EGFR are shown in blue. The green and orange alcohol groups represent the different coupling partners for the linker with either a 2-aminophenol or 3-aminophenol moiety (middle). The green and orange molecules represent the different attachment point of the linker (right).

**Figure 2:**

A. Graphical representation of the selectivity data of **2f** (left) and **3f** (right) assessed by DSF. The chemical structures are shown as well as a phylogenetic tree (Cell Signaling Technology) used to highlight ΔT_m data that were depicted as red circles as indicated in the figure capture. **B.** Selectivity profile of **3f** assessed by the KinomeScan panel at a screening concentration of 1 μ M. Mutants and atypical kinases are depicted in the lower panel. **C.** Waterfall plot of the selectivity data of **3f**. The remaining activity (%) was plotted against the number of tested kinases. Kinase mutants below 35% binding activity were highlighted by red dots, wild type kinases by blue dots. The selectivity score of **3f** was determined with a cut off value of 35% remaining activity. **D.** Table showing the top hits of the KINOMEScan profiling of **3f** with a remaining activity <35% at 1 μ M.

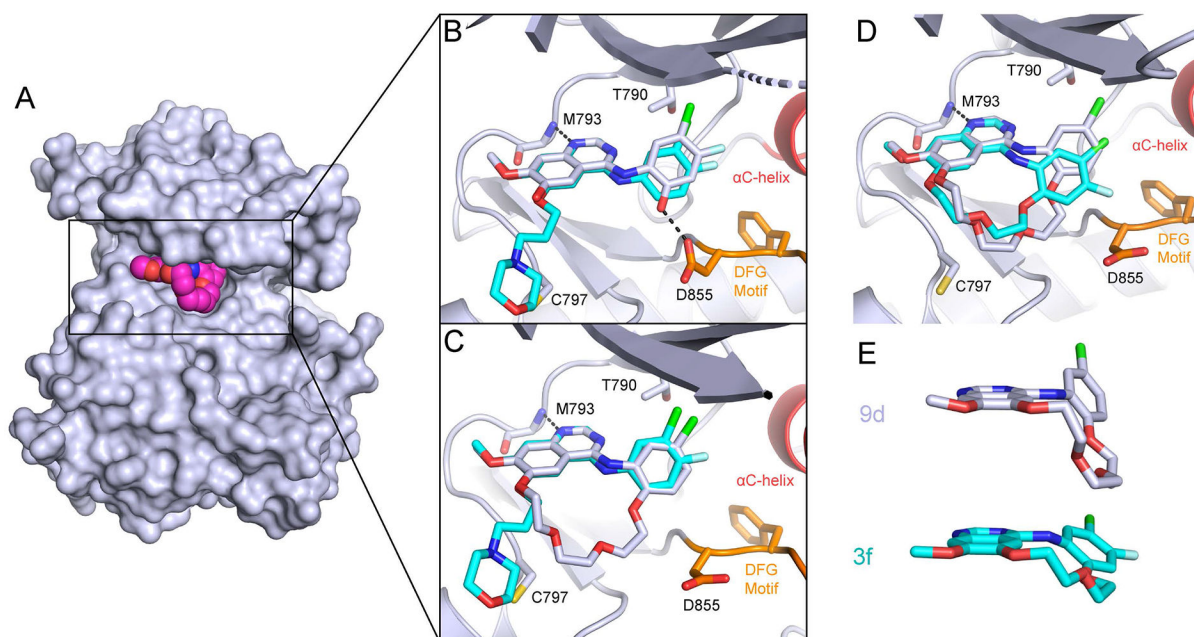


Figure 3.

A. Crystal structure of EGFR WT in complex with **9d** (PDB: 7U99). **B.** Superimposition of the structure of **2f** (PDB: 7U9A, grey carbon atoms) and gefitinib (PDB: 2ITZ, cyan carbon atoms). **C.** Superimposition of the structure of the macrocycle **9d** (grey carbon atoms) and gefitinib (cyan carbon atoms). **D.** Superimposition of the catalytic domain structure in complex with **9d** (grey carbon atoms) and **3f** (cyan carbon atoms) which was docked into EGFR WT (Glide, Schrodinger). **E.** Comparison of the different binding modes of the two macrocycles **9d** (grey) and **3f** (blue).

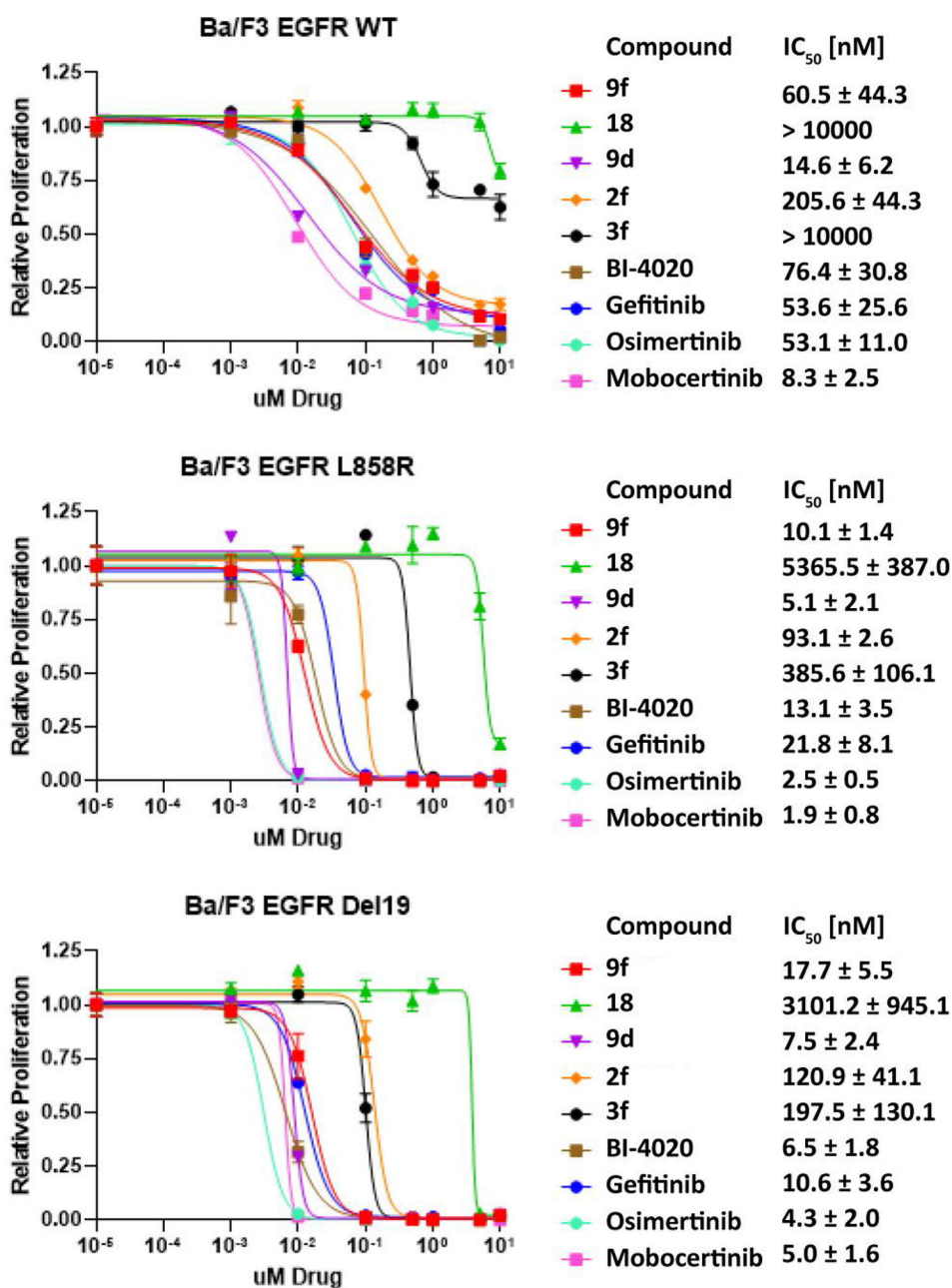
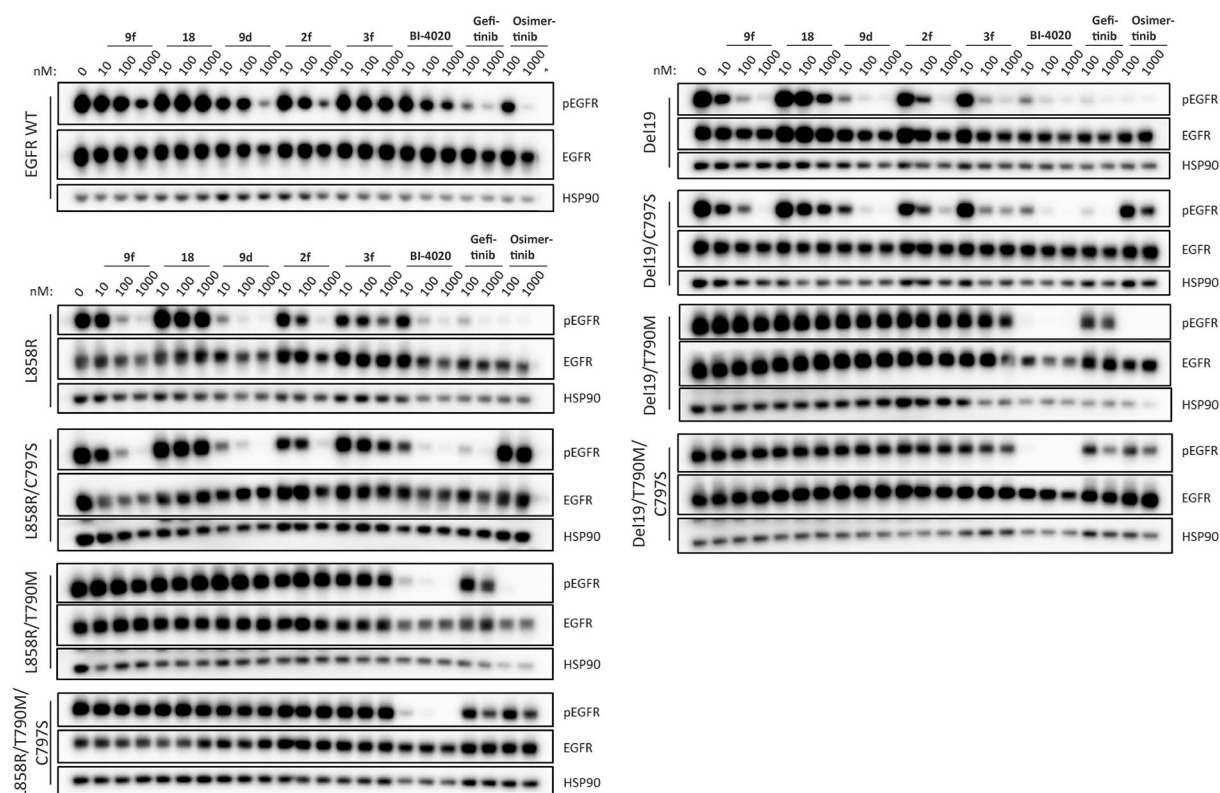
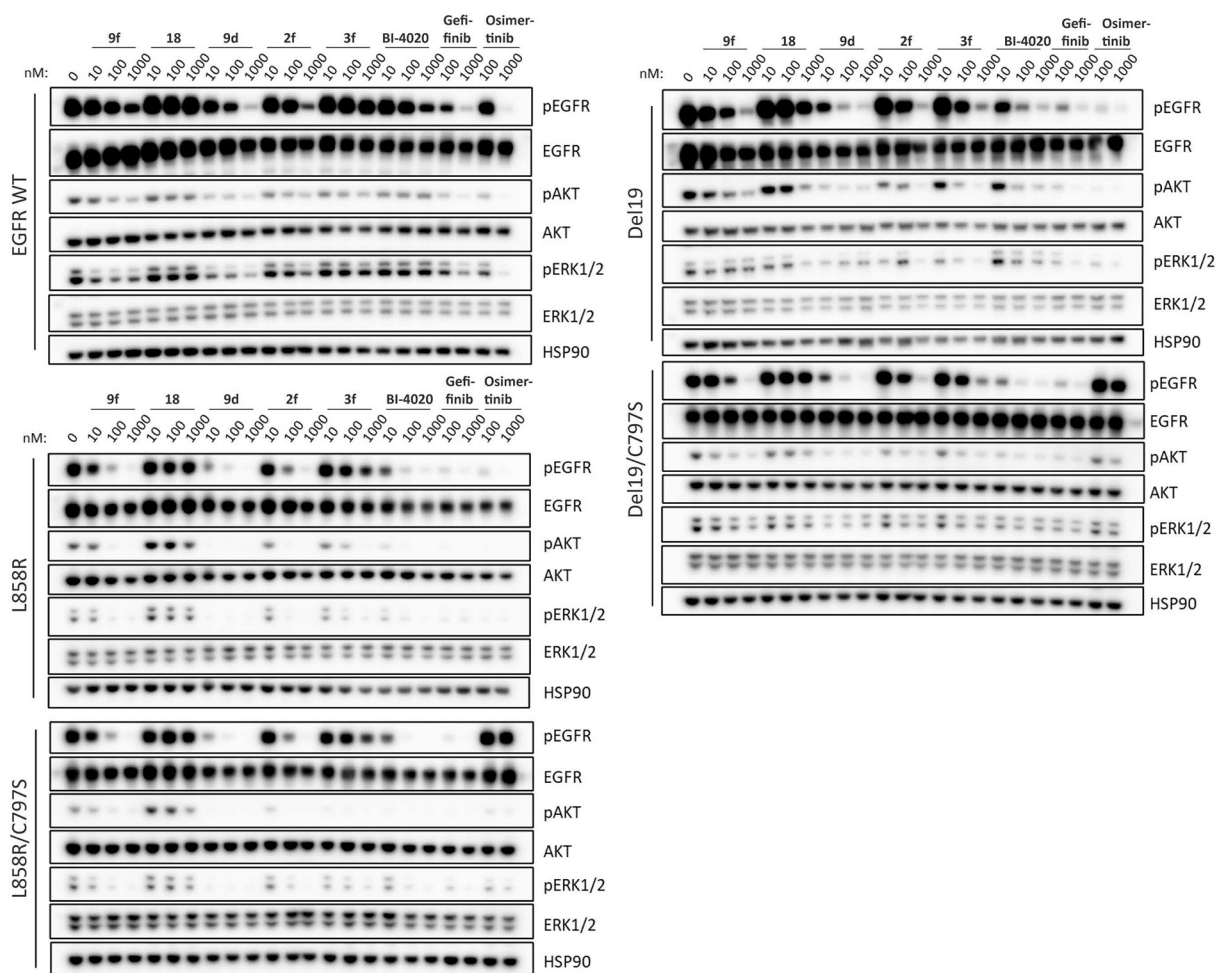


Figure 4:

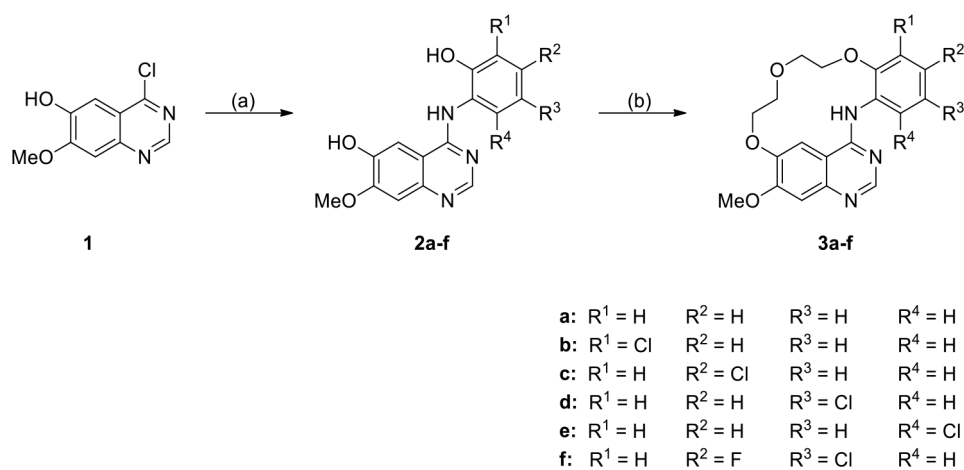
The dose-response curves of the selected compounds for determination of cellular IC₅₀ values. The respective IC₅₀ values and errors are given in the legend.

**Figure 5:**

Results of Western blot assays using compounds **9f**, **18**, **9d**, **2f**, **3f**, and BI-4020, gefitinib and osimertinib against EGFR WT and the most important EGFR mutants. EGFR phosphorylation (pEGFR) was used for the detection of pathway activation. Additionally, total protein levels of EGFR and HSP90 were monitored as loading and expression controls.

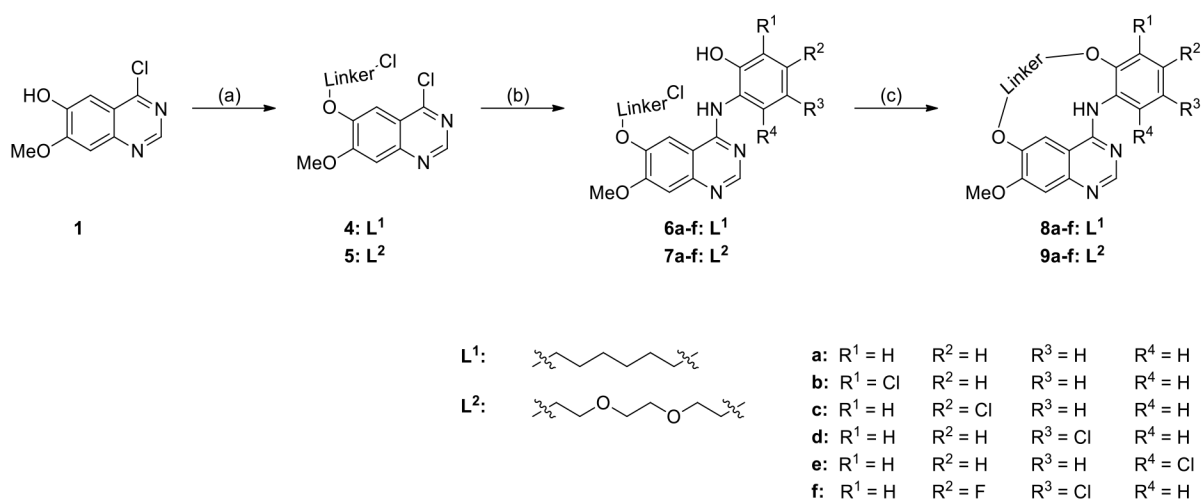
**Figure 6:**

Results of Western blot assays using compounds **9f**, **18**, **9d**, **2f**, **3f**, and BI-4020, gefitinib, and osimertinib against EGFR WT and the most interesting EGFR mutants L858R, L858R/C797S, Del19, and Del19/C797S. AKT and ERK1/2 phosphorylation (pAKT, pERK1/2) was used for the detection of signalling pathway inactivation. Additionally, total protein levels of AKT, ERK1/2, and HSP90 were monitored as loading and expression controls.



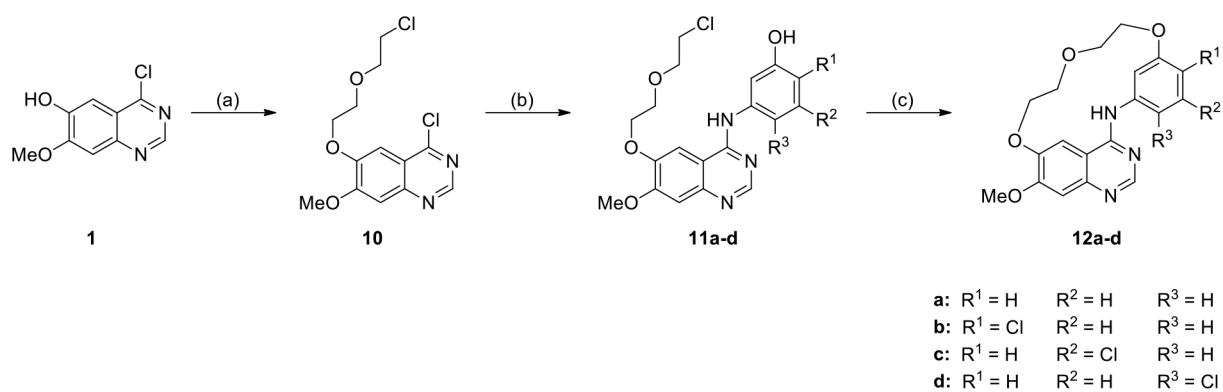
Scheme 1: Synthesis of macrocycles 3a-f.^a

^aReagents and conditions: (a) 2-aminophenols and ethanol, 17 h, 70 °C; (b) diethylene glycol, TPP, DIAD, THF and toluene, 20 h, rt - 40 °C.

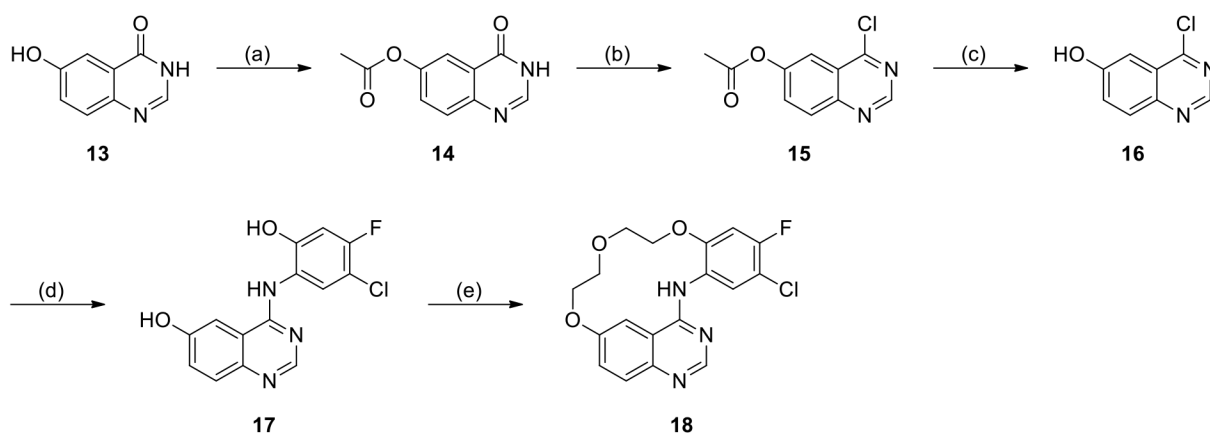


Scheme 2: Synthesis of the macrocycles 8a-f and 9a-f.^a

^aReagents and conditions: (a) 6-chloro-1-hexanol or 2-[2-(2-chloroethoxy)ethoxy]ethanol, TPP, DIAD and toluene, 2 - 5 h, rt - 80 °C; (b) 2-aminophenols and ethanol, 18 h, 70 °C; (c) NaH and DMF, 24 h, 0 - 60 °C.

**Scheme 3: Synthesis of the macrocycles 12a-d.^a**

^aReagents and conditions: (a) 2-(2-chloroethoxy)ethanol, TPP, DIAD and toluene, 2 h, rt - 80 °C; (b) 3-aminophenols and ethanol, 18 h, 70 °C; (c) NaH and DMF, 24 h, 0 - 60 °C.

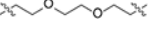
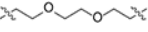
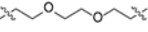
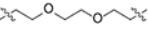
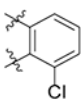
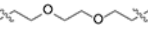
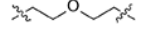
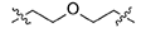
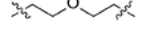
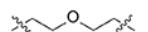
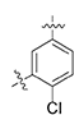
**Scheme 4: Synthesis of the macrocycle 18.^a**

^aReagents and conditions: (a) acetic anhydride and pyridine, 1 h, 50 °C; (b) thionyl chloride and DMF, 3 h, 70 °C; (c) NH₃ and methanol, 1 h, rt; (d) 2-amino-4-chloro-5-fluorophenol and ethanol, 18 h, 70 °C; (e) diethylene glycol, TPP, DIAD, THF and toluene, 20 h, rt - 40 °C.

Table 1:

Potency of the quinazoline-based macrocycles against EGFR WT and its mutants.

Compound ID	Linker	Aromat	Percent of control activity [%] ^a			IC ₅₀ [nM] ^b		
			WT	Del19	d747-752/ P753S	WT	L858R	L858R/ T790M
2f			1.9 ± 0.2	0.9 ± 0.1	0.4 ± 0.1	1795.3 ± 859.1	18.8 ± 2.2	1338.0 ± 631.5
18			55.2 ± 3.6	23.5 ± 0.7	60.4 ± 0.5	> 100000	> 100000	> 10000
3a			n.d.	n.d.	n.d.	> 100000	685.2 ± 59.2	1660.0 ± 235.5
3b			49.1 ± 3.9	14.6 ± 1.7	39.8 ± 0.3	> 100000	> 100000	> 10000
3c			80.4 ± 3.2	52.0 ± 0.9	76.2 ± 2.1	> 100000	> 100000	> 10000
3d			13.7 ± 2.9	1.1 ± 0.6	8.1 ± 1.7	> 100000	127.5 ± 30.1	4998.0 ± 706.4
3e			81.7 ± 8.1	64.9 ± 2.6	77.4 ± 1.4	n.d.	n.d.	n.d.
3f			44.5 ± 6.0	12.2 ± 0.2	40.5 ± 1.9	> 100000	820.8 ± 282.0	> 10000
8a			n.d.	n.d.	n.d.	> 100000	> 100000	> 10000
8b			62.2 ± 2.9	29.0 ± 2.3	53.8 ± 0.9	> 100000	> 100000	> 10000
8c			84.8 ± 0.6	93.1 ± 0.3	91.5 ± 6.4	> 100000	> 100000	> 10000
8d			14.9 ± 2.5	3.7 ± 0.9	10.6 ± 1.7	> 100000	738.6 ± 337.2	> 10000
8e			26.9 ± 3.3	7.2 ± 1.0	13.6 ± 0.9	> 100000	275.1 ± 22.7	3189 ± 114.2

Compound ID	Linker	Aromat	Percent of control activity [%] ^a			IC ₅₀ [nM] ^b		
			WT	Del19	d747-752/ P753S	WT	L858R	L858R/ T790M
8f			15.3 ± 2.3	9.9 ± 5.2	16.2 ± 3.6	> 100000	655.1 ± 294.6	> 10000
9a			n.d.	n.d.	n.d.	> 100000	87.6 ± 10.3	5312.0 ± 209.9
9b			48.0 ± 13.1	17.6 ± 1.1	29.1 ± 2.4	> 100000	448.5 ± 40.3	> 10000
9c			11.9 ± 0.6	5.2 ± 0.4	6.9 ± 0.4	> 100000	38.3 ± 5.5	1886 ± 41.5
9d			0.5 ± 0.5	0.0 ± 0.0	-0.2 ± 0.3	> 100000	0.7 ± 0.5	318.2 ± 8.6
9e			12.9 ± 0.5	4.6 ± 0.4	3.2 ± 0.8	> 100000	32.6 ± 2.8	1057.0 ± 34.3
9f			2.9 ± 0.0	0.3 ± 0.5	0.4 ± 0.9	> 100000	2.8 ± 1.6	734.7 ± 365.8
12a			n.d.	n.d.	n.d.	> 100000	> 100000	> 10000
12b			44.5 ± 3.3	29.0 ± 0.8	33.8 ± 0.1	> 100000	1119.0 ± 238.8	> 10000
12c			74.0 ± 1.9	70.9 ± 0.1	78.0 ± 1.5	> 100000	9680.0 ± 5258.9	> 10000
12d			95.1 ± 9.6	133.7 ± 40.9	92.2 ± 2.7	> 100000	19880.0 ± 3622.3	> 10000
Gefitinib			n.d.	n.d.	n.d.	> 100	3.8 ± 1.8	> 100
BI-4020			n.d.	n.d.	n.d.	> 100	5.0 ± 2.6	0.02 ± 0.01
Osimertinib			n.d.	n.d.	n.d.	34.5 ± 6.7	1.8 ± 0.3	0.4 ± 0.4

^aValues were determined at a screening concentration of 1 μM by Reaction Biology, using a radiometric protein kinase assay in duplicates.

^bIC₅₀ values were determined using a HTRF assay in a 12-point dose response curve (mean ± SD, n = 3 in triplicate).

Table 2:

Cell based activity data (Ba/F3 cell model) of the most interesting inhibitors tested against cells expressing EGFR WT and a number of mutants. IC₅₀ values for cell proliferation were assessed by a cell titer-glo assay in triplicates ± SD (Green indicates a high and red a low potency against the desired kinase).

Compound ID	WT	IC ₅₀ [nM]								Ex20ins. SVD	Ex20ins. NPG	Ex20ins. GY
		L858R	L858R/ C797S	L858R/ T790M	L858R/ T790M/ C797S	Del19	Del 19/ C797S	Del 19/ T790M	Del 19/ T790M/ C797S			
2f	205.6 ± 44.3	93.1 ± 2.6	141.4 ± 19.3	>1 x 10 ⁴	>1 x 10 ⁴	120.9 ± 41.1	119.6 ± 21.4	>1 x 10 ⁴	>1 x 10 ⁴	>1 x 10 ⁴	>1 x 10 ⁴	5184.7 ± 892.2
18	>1 x 10 ⁴	5365.5 ± 387.0	>1 x 10 ⁴	>1 x 10 ⁴	>1 x 10 ⁴	3101.2 ± 945.1	2762.8 ± 459.6	>1 x 10 ⁴	>1 x 10 ⁴	>1 x 10 ⁴	>1 x 10 ⁴	>1 x 10 ⁴
3f	>1 x 10 ⁴	385.6 ± 106.1	749.6 ± 124.9	>1 x 10 ⁴	>1 x 10 ⁴	197.5 ± 130.1	147.9 ± 13.2	>1 x 10 ⁴	>1 x 10 ⁴	>1 x 10 ⁴	>1 x 10 ⁴	>1 x 10 ⁴
9d	14.6 ± 6.2	5.1 ± 2.1	6.2 ± 1.7	>1 x 10 ⁴	>1 x 10 ⁴	7.5 ± 2.4	6.3 ± 2.0	6474.2 ± 1618.0	>1 x 10 ⁴	3163.1 ± 897.2	1545.0 ± 225.9	896.7 ± 307.5
9f	60.5 ± 26.3	10.1 ± 1.4	17.9 ± 4.9	>1 x 10 ⁴	>1 x 10 ⁴	17.7 ± 5.5	15.3 ± 2.4	>1 x 10 ⁴	>1 x 10 ⁴	>1 x 10 ⁴	4541.6 ± 1383.0	2535.8 ± 1129.5
Gefitinib	53.6 ± 25.6	21.8 ± 8.1	54.2 ± 21.6	>1 x 10 ⁴	>1 x 10 ⁴	10.6 ± 3.6	12.5 ± 2.3	>1 x 10 ⁴	>1 x 10 ⁴	6981.5 ± 1667.5	2779.2 ± 1107.9	1757.0 ± 442.8
BI-4020	76.4 ± 30.8	13.1 ± 3.5	11.1 ± 3.3	8.8 ± 7.5	3.4 ± 1.3	6.5 ± 1.8	2.3 ± 0.5	2.8 ± 0.9	2.8 ± 1.5	1412.1 ± 313.3	803.7 ± 366.8	1967.5 ± 1615.0
Osimertinib	53.1 ± 11.0	2.5 ± 0.5	1475.0 ± 129.6	10.1 ± 8.8	1409.5 ± 145.0	4.3 ± 2.0	1320.5 ± 135.0	4.5 ± 1.5	1503.9 ± 109.8	300.1 ± 74.0	60.6 ± 36.2	154.1 ± 40.0
Mobocertinib	8.3 ± 2.5	1.9 ± 0.8	1133.7 ± 823.5	27.1 ± 26.7	2168.1 ± 172.0	5.0 ± 1.6	1062.9 ± 277.4	8.9 ± 4.0	3814.5 ± 2299.0	47.6 ± 23.8	9.0 ± 2.7	10.1 ± 8.6